

Generalized Modeling and Fundamental Limits for Multiple-Access Integrated Sensing and Communication Systems

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Abstract—In this paper, we propose a generalized state-dependent channel modeling and present fundamental limits for multiple-access integrated sensing and communication (ISAC) systems. The model proposed extends the latest studies by Kobayashi *et al.* and Ahmadipour *et al.*, which explicitly accounts for more practical scenarios with correlated sensing and channel states, and imperfect channel state information at the receiver (CSIR). For the model considered, we devise an achievable scheme that combines message cooperation and joint compression of past transmitted codeword and echo signals (a form of strictly causal feedback) via distributed Wyner-Ziv coding at each user to realize “simultaneously cooperative communication and sensing”. The corresponding achievable rate-distortion region is derived and a numerical example is constructed to illustrate the potential gain of the proposed scheme. It is found that the compressed information sent is not only useful for further enhancing communication (particularly in the case without CSIR), but also helpful in improving the sensing performance of the transmitters.

I. INTRODUCTION

The integrated sensing and communication (ISAC), in which sensing and communication share the same frequency band and hardware, has emerged as a key technology in future 6G networks [1]. While substantial efforts have been devoted to investigating the key techniques for ISAC in different scenarios and system architectures, the optimal fundamental tradeoff between communication and sensing performance in such integrated systems remains largely unclear.

To tackle this challenge, pioneering studies [2]–[4] have exploited the state-dependent channel approach to model ISAC systems and proposed a notion of capacity-distortion tradeoff as the key performance metric from an information-theoretic point of view. Specifically, with the assumption of sensing state coinciding with channel state and with perfect channel state information at the receiver (CSIR), the authors investigate capacity-distortion tradeoff limits for ISAC over memoryless point-to-point channels, multiple access channels (MACs), and broadcast channels (BCs) in [2]–[4], respectively. In particular, for ISAC over MACs [3], the authors demonstrate that each transmitter can decode partial messages of the other transmitter, which can then be leveraged for both state sensing and message cooperation. An achievable scheme that mimics the block Markov encoding strategy of [5] for MACs with generalized output feedback is proposed. As a result, the rate-distortion region can be enlarged compared with the conventional resource sharing approach.

While the studies mentioned above provide important preliminary results on fundamental performance limits of ISAC,

Related works	Model assumption			Achievable scheme	
	Generalized Feedback	CSIR	Sensing states	Message cooperation	Sending compressed information
Willems [5]	✓			✓	×
Lapidoth and Steinberg [7] Li, Simeone, and Yener [8]	Strictly causal side information ($Z_k^{t-1} = S_k^{t-1}$, $k = 1, 2$)	ϕ	No sensing task	×	Compression of past state Joint compression of past state and past codeword
Kobayashi <i>et al.</i> [3]	✓	Explicit modeling with perfect CSIR ($S_R^n = (S_R^1, S_R^2)$)	Same as channel state ($S_R^n = S_k^n$)	✓	×
Ahmadipour <i>et al.</i> [9]	✓	Viewed as part of channel output	Same as channel state ($S_R^n = S_k^n$)	✓	Regular quantization; send the compressed information as part of common message
Our current work	✓	Explicit modeling with imperfect CSIR ($S_R^n: p(s_R, s)$)	Correlated $S_R^n: p(s_{T_1}, s_{T_2}, s)$	✓	Wyner-Ziv compression; send the compressed information as part of both common and private messages

Fig. 1. A summary of the key differences between our work and related studies on the system model and the key ingredient of the achievable scheme. the assumptions considered in [2]–[4] are rather restricted. In many practical ISAC scenarios such as connected vehicular networks, the sensing states and channel states are usually correlated but not the same. A intuitive example is that a base station may wish to estimate the position or speed of a moving car, rather than the full communication channel associated with the car. Moreover, the CSIR can hardly be perfect due to the channel estimation error. Given these practical constraints, how to fully utilize the echo signals and design proper encoding strategies for ISAC systems to enlarge the rate-distortion region still remains open.

In the special case without any CSIR and the echo signals are perfect state sequences (i.e., with perfect sensing), ISAC over MACs are reduced to a class of conventional state-dependent MACs with strictly causal state information at transmitters, as studied in [6]–[8]. In [6], a common state sequence is considered to be known strictly causally by the transmitters, while in [7], two independent state sequences are known strictly causally by two transmitters respectively. The results in [6], [7] demonstrate that transmitting the compressed state sequence to the decoder can enlarge the rate region. Considering the same model as in [7], the authors in [8] further show that by transmitting a jointly compressed sequence of codeword and channel states of the past block, an additional improvement can be obtained. Such results motivate us that transmitting additional compressed information of echo signals from each user in ISAC can also be useful to improve the system performance, particularly in the absence of CSIR.

In [9], CSIR is absorbed into the channel output, and the performance of ISAC over MACs is further enhanced by letting each transmitter use coding to convey information related to the echo signals to other transmitter. The authors

show that in addition to the message cooperation, sending quantized information related to the echo signals as part of the common messages that can be decoded by the other transmitter can further improve the sensing performance of transmitters.

In this paper, we consider a generalized channel modeling for multiple-access ISAC systems to explicitly account for the scenarios with correlated sensing and channel states, and imperfect CSIR. For the model considered, we further propose a new achievable scheme that combines the idea of message cooperation of [3], [5] and joint compression of past transmitted codeword and echo signals via distributed Wyner-Ziv coding, in similar spirit of [7]. In particular, the echo feedback-induced message cooperation is beneficial for both sensing and communication, which has been shown in [3]. The compressed information sent can also bring dual benefits. On the one hand, the common and private compressed information sent helps to convey additional information about the channel state to the receiver, which in turn uses such information to better perform coherent message decoding. On the other hand, each transmitter can extract the common compressed information sent by the other transmitter about the channel state through echo signals, which thus facilitates its state estimation. Compared with the regular quantization in [9], the proposed scheme via Wyner-Ziv coding can effectively reduce the amount of compressed information sent thus achieving better performance in both sensing and communication. Overall, such an achievable scheme leads to “simultaneously cooperative communication and sensing” between the transmitters in some sense.

The corresponding achievable rate-distortion region is derived. A numerical example is further constructed to demonstrate that our proposed scheme is potentially to achieve better performance in both sensing and communication compared with the state-of-the-art methods [3], [9]. A brief summary of the key differences between our work and related studies on system model and achievable scheme is provided in Fig. 1.

II. SYSTEM MODEL

Consider general ISAC over discrete memoryless (DM) MAC as shown in Fig. 2. Transmitter $k \in \{1, 2\}$ wishes to convey a message $W_k \in [1 : 2^{nR_k}] \triangleq \{1, 2, \dots, 2^{nR_k}\}$ to the receiver over a state-dependent DM MAC while simultaneously estimating the sensing state sequence $S^n_{T_k}$ via output feedback Z^n_k . Here output feedback models the communication echo signal reflected back to the transmitter.

The state-dependent DM MAC considered in Fig. 2 is denoted by $(\mathcal{X}_1 \times \mathcal{X}_2, \mathcal{S}, P_{Y Z_1 Z_2 | X_1 X_2 S}, \mathcal{Y}, \mathcal{Z}_1, \mathcal{Z}_2)$ with input alphabets $\mathcal{X}_1 \times \mathcal{X}_2$, state alphabet $\mathcal{S}$, output alphabet $\mathcal{Y}$, feedback alphabets $\mathcal{Z}_1 \times \mathcal{Z}_2$, and the collection of conditional probability mass functions $P_{Y Z_1 Z_2 | X_1 X_2 S}$. The channel state sequence S^n is assumed to be i.i.d. with $P_{S^n}(s^n) = \prod_{i=1}^n P_S(s_i)$. The channel is memoryless in the sense that at each time instance $i \in [1 : n]$, there is $P(y_i, z_{1,i}, z_{2,i} | x_1^n, x_2^n, s^n, y^{i-1}, z_1^{i-1}, z_2^{i-1}) = P(y_i, z_{1,i}, z_{2,i} | x_{1,i}, x_{2,i}, s_i)$.

In the ISAC model considered, the joint distribution of channel state S , CSIR S_R , and sensing states $S_{T_k}, k \in \{1, 2\}$ is given by $P_{S S_{T_1} S_{T_2} S_R}$. The CSIR sequence S^n_R is i.i.d. according to $P_{S^n_R}(s^n_R) = \prod_{i=1}^n P_{S_R}(s_{R,i})$ and the associated S_R is assumed to be correlated with channel state S but not necessarily the same. This modeling thus includes imperfect CSIR

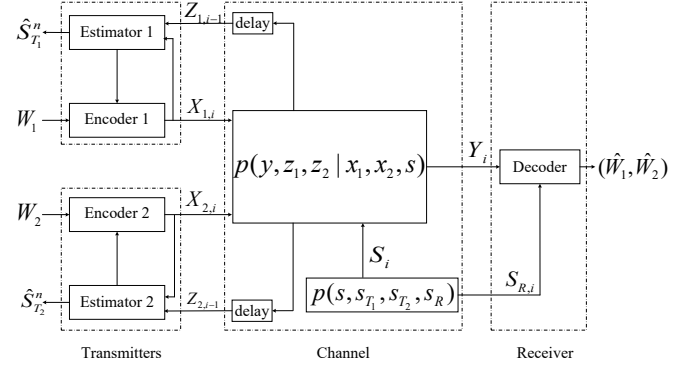


Fig. 2. ISAC over MAC with correlated sensing and channel states as well as imperfect CSIR.

($S_R \neq S$) in general and perfect CSIR ($S_R = S$) as a special case. Moreover, the sensing state sequence $S^n_{T_k}, k \in \{1, 2\}$ is also i.i.d. according to $P_{S^n_{T_k}}(s^n_{T_k}) = \prod_{i=1}^n P_{S_{T_k}}(s_{T_k,i})$, and the associated S_{T_k} is correlated with the channel state S .

For the model considered in Fig. 2, a $(2^{nR_1}, 2^{nR_2}, n)$ code consists of

- 1) two message set $\mathcal{W}_k = [1 : 2^{nR_k}], k \in \{1, 2\}$;
- 2) two encoders that encoder $k \in \{1, 2\}$ assigns a symbol $x_{k,i} = f_{k,i}(w_k, z_k^{i-1})$ to each message $w_k \in \mathcal{W}_k$ and delayed feedback $z_k^{i-1} \in \mathcal{Z}_k^{i-1}$;
- 3) a decoder that produces a message pair $(\hat{w}_1, \hat{w}_2) = h(y^n, s^n_R) \in \mathcal{W}_1 \times \mathcal{W}_2$ upon observing y^n and s^n_R ;
- 4) two state estimators that estimator $k \in \{1, 2\}$ assigns an estimated sensing sequence $\hat{s}^n_{T_k} \in \hat{\mathcal{S}}^n_{T_k}$ based on the codeword $x^n_k \in \mathcal{X}_k^n$ and receiving feedback $z^n_k \in \mathcal{Z}_k^n$.

Similar to the existing studies [3], [9], the sensing performance is measured by the expected distortion of the state estimated, i.e., $\mathbb{E}[d_k(S^n_{T_k}, \hat{s}^n_{T_k})] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[d_k(S_{T_k,i}, \hat{s}_{T_k,i})]$, where $d_k : S_{T_k} \times \hat{S}_{T_k} \rightarrow [0, \infty)$ is a bounded distortion function. A rate-distortion tuple (R_1, R_2, D_1, D_2) is said to be achievable if there exist a sequence of $(2^{nR_1}, 2^{nR_2}, n)$ codes with arbitrarily small error probability for decoding, i.e., $\lim_{n \rightarrow \infty} Pr((\hat{W}_1, \hat{W}_2) \neq (W_1, W_2)) = 0$, and the sensing distortion constraints $\limsup_{n \rightarrow \infty} \mathbb{E}[d_k(S^n_{T_k}, \hat{s}^n_{T_k})] \leq D_k, k \in 1, 2$ are satisfied. For any (D_1, D_2) , the capacity-distortion region $\mathcal{C}(D_1, D_2)$ is defined as the closure of achievable rate tuple (R_1, R_2) such that (R_1, R_2, D_1, D_2) is achievable.

III. MAIN RESULT

In this section, for the two-user ISAC over MAC with correlated sensing and channel states as well as imperfect CSIR, we propose a new achievable scheme that combines the idea of message cooperation and joint compression of past transmitted codeword and echo signals in which distributed Wyner-Ziv coding, block Markov encoding, and backward decoding are used. The main result of our new inner bound for capacity-distortion region is provided.

Consider the state estimator for transmitter $k \in \{1, 2\}$

$$\hat{s}_{T_k}(x_k, u_k, z_k, v_{k,c}) = \arg \min_{s'_{T_k} \in \mathcal{S}_{T_k}} \sum_{s_{T_k} \in \mathcal{S}_{T_k}} P_{S_{T_k} | X_k U_k Z_k V_{k,c}}(s_{T_k} | x_k u_k z_k v_{k,c}) d_k(s_{T_k}, s'_{T_k}), \quad (1)$$

where U_k denotes the union of previous common message parts of two transmitters and current common message part

$$\begin{aligned}
R_1 &\leq I(U_1; Z_2|UX_2) + I(X_1; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_2S_R) \\
&\quad + \min\{0, -I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1) + I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY)\}, \\
R_2 &\leq I(U_2; Z_1|UX_1) + I(X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_1S_R) \\
&\quad + \min\{0, -I(V_{2,c}V_{2,p}; UU_1U_2X_2Z_2) + I(V_{2,c}V_{2,p}; V_{1,c}V_{1,p}S_RY)\}, \\
R_1 + R_2 &\leq I(U_1; Z_2|UX_2) + I(U_2; Z_1|UX_1) + I(X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2S_R) \\
&\quad + \min\{0, -I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1) + I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY), \\
&\quad - I(V_{2,c}V_{2,p}; UU_1U_2X_2Z_2) + I(V_{2,c}V_{2,p}; V_{1,c}V_{1,p}S_RY), \\
&\quad - I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1) - I(V_{2,c}V_{2,p}; UU_1U_2X_2Z_2) + I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY) \\
&\quad + I(V_{2,c}V_{2,p}; V_{1,c}V_{1,p}S_RY) - I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}|S_RY)\}, \\
R_1 + R_2 &\leq I(U_1; Z_2|UX_2) + I(U_2; Z_1|UX_1) - I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1) - I(V_{2,c}V_{2,p}; UU_1U_2X_2Z_2) \\
&\quad + I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY) + I(V_{2,c}V_{2,p}; V_{1,c}V_{1,p}S_RY) - I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}|S_RY) \\
&\quad + I(X_1; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_2S_R) + I(X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_1S_R), \\
R_1 + R_2 &\leq I(UU_1U_2X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|S_R) \\
&\quad + \min\{0, -I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1) + I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY), \\
&\quad - I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1) + I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY), \\
&\quad - I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1) - I(V_{2,c}V_{2,p}; UU_1U_2X_2Z_2) + I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY) \\
&\quad + I(V_{2,c}V_{2,p}; V_{1,c}V_{1,p}S_RY) - I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}|S_RY)\} \tag{2}
\end{aligned}$$

of transmitter k , $V_{k,c}$ denotes the common part of compressed information of transmitter k , and $\bar{k} \triangleq 3 - k$. Based on (1), we establish an achievable rate-distortion region as follows.

Theorem 1: Let (D_1, D_2) be the given distortion constraints for sensing states. For two-user ISAC over MAC with correlated sensing and channel states as well as imperfect CSIR, an achievable rate-distortion region is given in (2) on the top of the current page where $\mathbb{E}[d_k(S_{T_k}^n, S_{T_k}^n)] \leq D_k, k \in \{1, 2\}$, and the joint distribution of all variables is given by

$$\begin{aligned}
&P_U P_{U_1|U} P_{U_2|U} P_{X_1|U_1} P_{X_2|U_2} P_{SS_R S_{T_1} S_{T_2}} P_{Y Z_1 Z_2|X_1 X_2 S} \\
&P_{V_{1,c}V_{1,p}|UU_1U_2X_1Z_1} P_{V_{2,c}V_{2,p}|UU_1U_2X_2Z_2} \\
&P_{\hat{S}_{T_1}|X_1Z_1U_2V_{2,c}} P_{\hat{S}_{T_2}|X_2Z_2U_1V_{1,c}}. \tag{3}
\end{aligned}$$

Remark 1: Based on the result in Theorem 1, we make the following remarks.

- 1) Theorem 1 contains the result in [3] as a special case without transmitting the compressed sequences, i.e., $V_1V_2 = \phi$ for the case that the sensing state is the same as the channel state and the receiver has perfect CSI.
- 2) When the generalized feedback is assumed to be coincided with the channel state $Z_1 = S_1, Z_2 = S_2$ and no sensing tasks are required at the transmitters, the rate-distortion region provided in Theorem 1 specialized to the result in [7] and [8] by sending the compressed sequences of channel states and the joint compressed sequences of channel states and codewords, respectively.

IV. ACHIEVABILITY PROOF

The achievable scheme proposed combined with distributed Wyner-Ziv coding, block Markov encoding, and backward decoding. The proposed coded transmission consists of $B + \bar{B}$ blocks. The last $\bar{B}$ blocks called ‘‘termination blocks’’ are necessary to guarantee that the receiver can successfully obtain the compressed information corresponding to block B . The discussion of termination blocks is provided in Appendix C, and in the section, we are focused on communication and sensing in the first B blocks. Each of the first B blocks is of length N channel uses. Transmitter $k \in \{1, 2\}$ transmits $B - 1$ i.i.d. messages $\{w_k^{(b)}\}_{b=1}^{B-1}$ over $B + \bar{B}$ blocks. Each message

$w_k^{(b)}$ is partitioned into two messages $\{w_{k,p}^{(b)}, w_{k,c}^{(b)}\}$. The subscripts ‘‘p’’ and ‘‘c’’ stand for ‘‘private message’’ and ‘‘common message’’, respectively. The messages $w_{k,p}^{(b)} \in [1 : 2^{NR_{k,p}}]$ and $w_{k,c}^{(b)} \in [1 : 2^{NR_{k,c}}]$ are uniformly distributed and mutually independent for all $k \in \{1, 2\}$ and $b \in [1 : B - 1]$.

A. Codebook Generation

Pick non-negative real numbers $R_{V_{1,c}}, R_{V_{1,p}}, R_{V_{2,c}}, R_{V_{2,p}}, R_{Z_{1,c}}, R_{Z_{1,p}}, R_{Z_{2,c}}$, and $R_{Z_{2,p}}$ satisfying

$$\begin{aligned}
R_{V_{1,c}} &\leq R_{Z_{1,c}}, R_{V_{1,p}} \leq R_{Z_{1,p}}, \\
R_{V_{2,c}} &\leq R_{Z_{2,c}}, R_{V_{2,p}} \leq R_{Z_{2,p}} \tag{4}
\end{aligned}$$

in a fashion that will be specified later. For each block $b \in [1 : B]$, the codebook is constructed as follows.

1) Generate $2^{NR_{Z_{1,c}}}$ sequences $v_{1,c}^N(\Delta_{1,c})$, $\Delta_{1,c} \in [1 : 2^{NR_{Z_{1,c}}}]$, i.i.d. according to $P_{V_{1,c}}(\cdot)$. Randomly partition the indices $\{\Delta_{1,c} : \Delta_{1,c} \in [1 : 2^{NR_{Z_{1,c}}}]\}$ into $2^{NR_{V_{1,c}}}$ bins. Denote by $\eta_{1,c}(\Delta_{1,c})$ the index of the bin to which $\Delta_{1,c}$ belongs, and by $\alpha_{1,c}(\eta_{1,c})$ the set of sequence indices corresponding to the bin number $\eta_{1,c}$. Similarly, generate $2^{NR_{Z_{1,p}}}$, $2^{NR_{Z_{2,c}}}$, $2^{NR_{Z_{2,p}}}$ sequences $v_{1,p}^N(\Delta_{1,p})$, $v_{2,c}^N(\Delta_{2,c})$, $v_{2,p}^N(\Delta_{2,p})$, respectively. Randomly partition the indices $\{\Delta_{1,p} : \Delta_{1,p} \in [1 : 2^{NR_{Z_{1,p}}}]\}$, $\{\Delta_{2,c} : \Delta_{2,c} \in [1 : 2^{NR_{Z_{2,c}}}]\}$, and $\{\Delta_{2,p} : \Delta_{2,p} \in [1 : 2^{NR_{Z_{2,p}}}]\}$ into $2^{NR_{V_{1,p}}}$, $2^{NR_{V_{2,c}}}$, and $2^{NR_{V_{2,p}}}$ bins. Denote by $\eta_{1,p}(\Delta_{1,p})$, $\eta_{2,c}(\Delta_{2,c})$, and $\eta_{2,p}(\Delta_{2,p})$ the indices of the bin to which $\Delta_{1,p}$, $\Delta_{2,c}$, and $\Delta_{2,p}$ belong, and by $\alpha_{1,p}(\eta_{1,p})$, $\alpha_{2,c}(\eta_{2,c})$, $\alpha_{2,p}(\eta_{2,p})$ the set of sequence indices corresponding to the bin number $\eta_{1,p}$, $\eta_{2,c}$, $\eta_{2,p}$, respectively.

2) Generate $2^{N(R_{1,c}+R_{V_{1,c}}+R_{2,c}+R_{V_{2,c}})}$ sequences $u^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1})$, $j_{b-1} \in [1 : 2^{NR_{1,c}}]$, $l_{b-1} \in [1 : 2^{NR_{V_{1,c}}}]$, $k_{b-1} \in [1 : 2^{NR_{2,c}}]$, $m_{b-1} \in [1 : 2^{NR_{V_{2,c}}}]$, i.i.d. according to $P_U(\cdot)$.

3) For each indices tuple $(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1})$, generate $2^{N(R_{1,c}+R_{V_{1,c}})}$ sequences $u_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b)$, $j_b \in [1 : 2^{NR_{1,c}}]$, $l_b \in [1 : 2^{NR_{V_{1,c}}}]$ i.i.d. according to $P_{U_1|U}(\cdot)$. Similarly, generate $u_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b)$, $k_b \in [1 : 2^{NR_{2,c}}]$, $m_b \in [1 : 2^{NR_{V_{2,c}}}]$.

4) For each indices tuple $(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b)$, generate $2^{N(R_{1,p} + R_{V_{1,p}})}$ sequences $x_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b, r_b, \beta_{1,b})$, $r_b \in [1 : 2^{NR_{1,p}}]$, $\beta_{1,b} \in [1 : 2^{NR_{V_{1,p}}}]$, i.i.d. according to $P_{X_1|U_1}(\cdot)$. Similarly, generate $2^{N(R_{2,p} + R_{V_{2,p}})}$ sequences $x_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b, s_b, \beta_{2,b})$, $s_b \in [1 : 2^{NR_{2,p}}]$, $\beta_{2,b} \in [1 : 2^{NR_{V_{2,p}}}]$.

B. Encoding

We set $j_0 = k_0 = l_0 = m_0 = 1$, $l_1 = m_1 = \beta_{1,1} = \beta_{2,1} = 1$, and $j_B = k_B = r_B = s_B = 1$. The operations of two transmitters in the first B blocks are shown as follows.

1) **Block 1.** Two transmitters send $x_1^N(1, 1, 1, 1, j_1, 1, r_1, 1)$ and $x_2^N(1, 1, 1, 1, k_1, 1, s_1, 1)$.

2) **Block $b \in [2 : B]$.** At the beginning of block b , transmitter 1 is cognizant of generalized feedback $z_1^N(b-1)$ and finds two indices k_{b-1}, m_{b-1} such that

$$\left(x_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}, r_{b-1}, \beta_{1,b-1}), \right. \\ z_1^N(b-1), u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), \\ u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), \\ \left. u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}) \right) \in \mathcal{T}_\epsilon^N. \quad (5)$$

Once obtaining k_{b-1}, m_{b-1} , transmitter 1 finds two indices $\Delta_{1,c,b}$ and $\Delta_{1,p,b}$ such that

$$\left(x_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}, r_{b-1}, \beta_{1,b-1}), \right. \\ v_{1,c}^N(\Delta_{1,c,b}), u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), \\ v_{1,p}^N(\Delta_{1,p,b}), u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}), \\ \left. z_1^N(b-1), u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}) \right) \in \mathcal{T}_\epsilon^N. \quad (6)$$

Denote by l_b and $\beta_{1,b}$ the bin number to which the indices $\Delta_{1,c,b}$ and $\Delta_{1,p,b}$ belong, respectively. Transmitter 1 sends $x_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b, r_b, \beta_{1,b})$ in block b . Similarly, transmitter 2 finds two indices j_{b-1}, l_{b-1} and sends $x_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b, s_b, \beta_{2,b})$.

C. Decoding

Decoding begins at the last block and proceeds backward.

1) **Termination blocks.** We assume that in the termination blocks, the decoder can successfully find the indices $l_B, m_B, \beta_{1,B}, \beta_{2,B}, l_{B+1}, m_{B+1}, \beta_{1,B+1}, \beta_{2,B+1}$, which is guaranteed by Appendix C.

2) **Block $b \in [B : 1]$.** The decoder has tuple $(l_{b+1}, m_{b+1}, \beta_{1,b+1}, \beta_{2,b+1})$, CSIR $s_R^N(b)$, and the channel output $y^N(b)$. It looks for $v_{1,c}^N(\Delta_{1,c,b+1}) \in \alpha_{1,c}(l_{b+1})$, $v_{1,p}^N(\Delta_{1,p,b+1}) \in \alpha_{1,p}(\beta_{1,b+1})$, $v_{2,c}^N(\Delta_{2,c,b+1}) \in \alpha_{2,c}(m_{b+1})$, and $v_{2,p}^N(\Delta_{2,p,b+1}) \in \alpha_{2,p}(\beta_{2,b+1})$ such that

$$\left(v_{1,c}^N(\Delta_{1,c,b+1}), v_{1,p}^N(\Delta_{1,p,b+1}), v_{2,c}^N(\Delta_{2,c,b+1}), \right. \\ \left. v_{2,p}^N(\Delta_{2,p,b+1}), s_R^N(b), y^N(b) \right) \in \mathcal{T}_\epsilon^N. \quad (7)$$

The indices $\Delta_{1,c,b+1}$, $\Delta_{1,p,b+1}$, $\Delta_{2,c,b+1}$, and $\Delta_{2,p,b+1}$ facilitates the decoding of the messages and the com-

pressed state sequence. Assuming (j_b, l_b, k_b, m_b) is decoded correctly in block $b+1$, the decoder finds $(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r_b, \beta_{1,b}, s_b, \beta_{2,b})$ such that

$$\left(v_{1,c}^N(\Delta_{1,c,b+1}), u_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b), \right. \\ v_{1,p}^N(\Delta_{1,p,b+1}), u_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b), \\ v_{2,c}^N(\Delta_{2,c,b+1}), x_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b, r_b, \beta_{1,b}), \\ v_{2,p}^N(\Delta_{2,p,b+1}), x_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b, s_b, \beta_{2,b}), \\ \left. s_R^N(b), y^N(b), u^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}) \right) \in \mathcal{T}_\epsilon^N. \quad (8)$$

The decoder repeats this step for block B to 1. The decoder output is $(j_b, r_b, k_b, s_b), b \in [1 : B-1]$.

D. State Estimation

At the beginning of block $b \in [2 : B+1]$, transmitter 1 has generalized feedback $z_1^N(b-1)$. After obtaining the index m_{b-1} , transmitter 1 looks for $v_{2,c}^N(\Delta_{2,c,b-1}) \in \alpha_{2,c}(m_{b-1})$ such that

$$\left(x_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}, r_{b-1}, \beta_{1,b-1}), \right. \\ z_1^N(b-1), v_{2,c}^N(\Delta_{2,c,b-1}), u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), \\ u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), \\ \left. u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}) \right) \in \mathcal{T}_\epsilon^N. \quad (9)$$

The compressed sequence $v_{2,c}^N(\Delta_{2,c,b-1})$ facilitates the state estimation. For block $b \in [1 : B-1]$, transmitter 1 outputs the estimated state sequence

$$\hat{s}_{T_1}(b) = \hat{s}_{T_1}(x_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b, r_b, \beta_{1,b}), u_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b), z_1^N(b), v_{2,c}^N(\Delta_{2,c,b+1})), \quad (10)$$

where the state estimator $\hat{s}_{T_1}(x_1, u_2, z_1, v_{2,c})$ is given in (1).

Similarly, for block $b \in [1 : B-1]$, transmitter 2 outputs the estimated state sequence

$$\hat{s}_{T_2}(b) = \hat{s}_{T_2}(x_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b, s_b, \beta_{2,b}), u_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b), z_2^N(b), v_{1,c}^N(\Delta_{1,c,b+1})). \quad (11)$$

E. Analysis of Error Probability and Expected Distortion

It can be proved that by letting $B, N \rightarrow \infty$, there is $P_e^N \rightarrow 0$ if the following conditions hold:

$$\begin{aligned} R_{1,c} + R_{V_{1,c}} &\leq I(U_1; Z_2|UX_2), \\ R_{2,c} + R_{V_{2,c}} &\leq I(U_2; Z_1|UX_1), \\ R_{z_{1,c}} + R_{z_{1,p}} &\geq I(V_{1,c}V_{1,p}; UU_1U_2X_1Z_1), \\ R_{z_{2,c}} + R_{z_{2,p}} &\geq I(V_{2,c}V_{2,p}; UU_1U_2X_2Z_2), \\ R_{z_{1,c}} - R_{V_{1,c}} + R_{z_{1,p}} - R_{V_{1,p}} &\leq I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY), \\ R_{z_{2,c}} - R_{V_{2,c}} + R_{z_{2,p}} - R_{V_{2,p}} &\leq I(V_{2,c}V_{2,p}; V_{1,c}V_{1,p}S_RY), \\ R_{z_{1,c}} - R_{V_{1,c}} + R_{z_{1,p}} - R_{V_{1,p}} + R_{z_{2,c}} - R_{V_{2,c}} \\ &\quad + R_{z_{2,p}} - R_{V_{2,p}} \leq I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}S_RY) \\ &\quad + I(V_{2,c}V_{2,p}; V_{1,c}V_{1,p}S_RY) - I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}|S_RY), \\ R_{1,p} + R_{V_{1,p}} &\leq I(X_1; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_2S_R), \\ R_{2,p} + R_{V_{2,p}} &\leq I(X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_1S_R), \\ R_{1,p} + R_{2,p} + R_{V_{1,p}} + R_{V_{2,p}} \\ &\leq I(X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2S_R), \end{aligned}$$

$$\begin{aligned}
R_{1,c} + R_{2,c} + R_{V_{1,c}} + R_{V_{2,c}} + R_{1,p} + R_{2,p} + R_{V_{1,p}} \\
+ R_{V_{2,p}} &\leq I(UU_1U_2X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|S_R), \\
R_{z_{1,c}} - R_{V_{1,c}} &\leq I(V_{1,c}; UU_1U_2X_2Z_2), \\
R_{z_{2,c}} - R_{V_{2,c}} &\leq I(V_{2,c}; UU_1U_2X_1Z_1).
\end{aligned} \tag{12}$$

Defining $R_1 = R_{1,p} + R_{1,c}$, $R_2 = R_{2,p} + R_{2,c}$. By Fourier-Motzkin elimination approach, one can eliminate auxiliary variables and obtain the achievable region as stated in Theorem 1. The analysis is provided in Appendix A. The corresponding analysis of distortion is provided in Appendix B.

V. EXAMPLE

The result in [3] is a special case of both our work and the result in [9]. Thus, we compare our achievable rate-distortion region with that in [9] via the following example.

Consider a MAC where the inputs X_1 and X_2 are binary. The channel output $Y = (Y_1, Y_2)$ is a binary tuple. The channel law is $Y_1 = X_1 \oplus S_1$, $Y_2 = X_2 \oplus S_{21} \oplus S_{22}$, where the channel states S_1, S_{21}, S_{22} are i.i.d. binary random variables. The transmitter 1's feedback $Z_1 = (Z_{11}, Z_{12})$ is a binary tuple, where $Z_{11} = X_1 \oplus S_{21}$, $Z_{12} = X_1 \oplus S_{22}$, and the transmitter 2's feedback $Z_2 = X_1 \oplus X_2 \oplus S_{21} \oplus N_{fb}$ is binary, where N_{fb} is a binary random variable. The receiver is assumed to have no CSI, i.e., $S_R = \phi$. The sensing states of two transmitters are assumed as $S_{T_1} = S_{T_2} = S_{21}$, and the Hamming distortion is considered as the distortion metric, i.e., $d(s_T, \hat{s}_T) = s_T \oplus \hat{s}_T$.

In this example, transmitter 1 can always achieve zero distortion. We focus on the achievable rate-distortion region (D_2, R_2) with the quality of service (QoS) constraint $R_1 \geq \alpha$. Specially, we use $p_1 * p_2$ to denote the convolution operation of two Bernoulli distributions with parameters p_1 and p_2 , i.e., $p_1 * p_2 = p_1(1 - p_2) + (1 - p_1)p_2$.

To characterize the achievable rate-distortion region induced by our achievable scheme, we consider

$$X_k = U_k \oplus \Theta_k = U \oplus \Sigma_k \oplus \Theta_k, \quad k \in \{1, 2\}, \tag{13}$$

where $U, \Sigma_1, \Sigma_2, \Theta_1, \Theta_2$ are independent binary random variables. Specially, we notice that $\Sigma_2 = 0$ is deterministic since the transmitter 1's feedback provides no information about X_2 . The compression variables are chosen as

$$\begin{aligned}
V_{1,c} &= \tilde{S}_{21} \oplus \tilde{S}_{22}, \quad V_{1,p} = N_{1,1} \oplus N_{2,1}, \\
V_{2,c} &= 0, \quad V_{2,p} = 0,
\end{aligned} \tag{14}$$

where $\tilde{S}_{2k}$ is the quantization of S_{2k} that satisfies $S_{2k} = \tilde{S}_{2k} \oplus N_{k,1} \oplus N_{k,0}$, $k \in \{1, 2\}$, where $N_{k,e} \sim \text{Bern}(p_{N_{k,e}})$, $e \in \{0, 1\}$ is the quantization variable. For this choice, transmitter 1 conveys information about S_{21} and $S_{21} \oplus S_{22}$ to transmitter 2 and receiver, respectively. The calculation of terms in the achievable scheme is presented in Appendix D and E.

Similarly, to characterize the achievable rate-distortion region induced by scheme proposed in [9], we still consider (13) where $U, \Sigma_1, \Sigma_2 = 0, \Theta_1, \Theta_2$ are independent binary random variables. The compression variables are chosen as $V_1 = \tilde{S}_{21} \oplus \tilde{S}_{22}$, $V_2 = 0$, where $\tilde{S}_{2k}$ is the quantization of S_{2k} that satisfies $S_{2k} = \tilde{S}_{2k} \oplus N_k$, and $N_k \sim \text{Bern}(p_{N_k})$ is the quantization variable. For this choice, transmitter 1 conveys information about S_{21} and $S_{21} \oplus S_{22}$ to transmitter 2 and receiver, respectively. In fact, when $N_1 = 0, N_2 = S_2$, transmitter 1 sends perfect S_1 to transmitter 2, which helps transmitter 2 to achieve a minimum sensing distortion. When

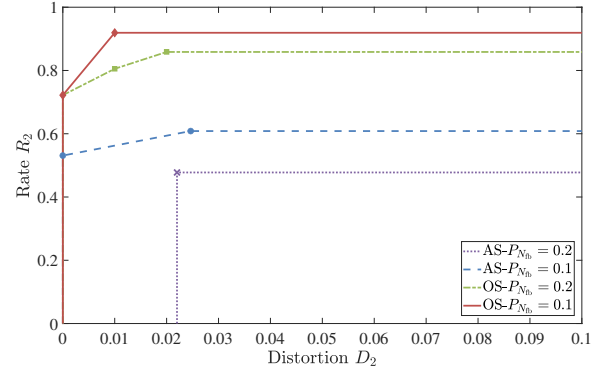


Fig. 3. Achievable rate-distortion region of transmitter 2.

$N_1 = 0, N_2 = 0$, transmitter 1 sends perfect $S_1 \oplus S_2$, which is beneficial for receiver to decode the message of transmitter 2. The calculation of terms is presented in Appendix F and G.

For $p_{S_1} = 0, p_{S_{21}} = 0.1, p_{S_{22}} = 0.1, \alpha = 0.4$, we evaluate rate-distortion region achieved by Theorem 1 and the result in [9] for $p_{N_{fb}} = 0.1$ and $p_{N_{fb}} = 0.2$, which is shown in Fig. 3, where AS denotes scheme proposed in [9], and OS denotes our proposed scheme. It can be found that our proposed scheme outperforms the achievable scheme in [9] for the cases considered. In particular, when $p_{N_{fb}} = 0.2$, the rate-distortion tradeoff of AS degrades into a point. The reason is that N_{fb} is part of noise of Z_2 . For a low noise power, i.e., $p_{N_{fb}} = 0.1$, transmitter 1 can send the perfect S_{21} as part of the common message to achieve zero distortion for transmitter 2 while satisfying its QoS constraint. When communication is also considered, transmitter 1 also transmit part information of S_{22} . Such coding can improve the achievable rate of transmit 2 while leading to a worse sensing performance. When $p_{N_{fb}} = 0.2$, transmitter 1 can only send the quantization of S_{21} while satisfying its QoS constraint. In this case, transmitter 2 cannot achieve zero distortion, and there remains no rate to support the transmission of the information of S_{22} . In our scheme, due to the joint consideration of message cooperation and compression information via Wyner-Ziv coding, for both two cases, transmitter 1 can always send perfect S_{21} for pure sensing and $S_{21} \oplus \tilde{S}_{22}$ for data transmission.

VI. CONCLUSION

In this work, we have investigated the fundamental limits of ISAC over memoryless MACs with correlated sensing and channel states, and imperfect CSIR. A new achievable scheme that combines message cooperation and joint compression of past transmitted codewords and echo signals via Wyner-Ziv coding is proposed, and the corresponding inner bound of the capacity-distortion region is derived. Via the numerical example constructed, we demonstrate that by jointly considering message cooperation and sending additional compressed information, two transmitters can cooperate to attain better sensing performance, and the decoder can support larger communication rates in general. It is finally remarked that while the inner bound established here improves upon that with message cooperation only, it is still sub-optimal in general for the ISAC over MACs considered.

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APPENDIX A ANALYSIS OF ERROR PROBABILITY

We focus on the decoding of the first B blocks assuming that the indices $l_B, m_B, \beta_{1,B}, \beta_{2,B}, l_{B+1}, m_{B+1}, \beta_{1,B+1}, \beta_{2,B+1}$ are decoded correctly. Without loss of generality, we assume that a specific sequence of $(l_1^{B+1}, m_1^{B+1}, \beta_{1,1}^{B+1}, \beta_{2,1}^{B+1})$ is chosen, and that $(j_b, k_b, r_b, s_b) = (1, 1, 1, 1), \forall b \in [1 : B]$.

Define the events in (15) and (16) on the top of next two pages. We can bound the average error probability as

$$\begin{aligned}
 P_e^{(n)} &\leq P(\mathcal{A}) + P(\mathcal{B}^c | \mathcal{A}^c) \\
 &\quad + P(\mathcal{G}_B | \mathcal{B}, \mathcal{A}^c) + P(\mathcal{H}_B | \mathcal{G}_B^c, \mathcal{B}, \mathcal{A}^c) \\
 &\quad + \sum_{b=2}^{B-1} (P(\mathcal{G}_b | \mathcal{H}_{b+1}^c, \mathcal{B}, \mathcal{A}^c) + P(\mathcal{H}_b | \mathcal{G}_b^c, \mathcal{H}_{b+1}^c, \mathcal{B}, \mathcal{A}^c)) \\
 &\quad + P(\mathcal{G}_1 | \mathcal{H}_2^c, \mathcal{B}, \mathcal{A}^c) + P(\mathcal{H}_1 | \mathcal{G}_1^c, \mathcal{H}_2^c, \mathcal{B}, \mathcal{A}^c) \\
 &\quad + P(\mathcal{M} | \mathcal{B}, \mathcal{A}^c). \tag{17}
 \end{aligned}$$

The first term in R.H.S. of (17) represents the error probability of estimating the common messages and compressed sequences during block 2 to B . By the law of large numbers, as $N \rightarrow \infty$, there is $P(\mathcal{A}_{1,b}^c(k_{b-1}, m_{b-1})) \rightarrow 0$ and $P(\mathcal{A}_{2,b}^c(j_{b-1}, l_{b-1})) \rightarrow 0$. For $(k, m) \neq (k_{b-1}, m_{b-1})$, there is

$$\begin{aligned}
 P(\mathcal{A}_{1,b}(k, m)) &= \sum_{u_2^N \in \mathcal{T}_\epsilon^N(U_2 | u_1^N u_1^N x_1^N z_1^N)} P_{U_2 | U}(u_2^N | u^N) \\
 &\stackrel{(a)}{\leq} 2^{-N(H(U_2 | U u_1^N x_1^N z_1^N) + \delta)} 2^{-N(H(U_2 | U) - \delta)} \\
 &= 2^{-N(I(U_2; U_1 X_1 Z_1 | U) - 2\delta)}
 \end{aligned}$$

$$\begin{aligned}
 &\stackrel{(b)}{=} 2^{-N(I(U_2; Z_1 | U U_1 X_1) - 2\delta)} \\
 &\stackrel{(c)}{=} 2^{-N(I(U_2; Z_1 | U X_1) - 2\delta)}, \tag{18}
 \end{aligned}$$

where (a) follows by the joint typicality lemma, (b) follows from the Markov chain $U_1 X_1 - U - U_2$, and (c) follows from the Markov chain $U_1 - U - U_2$. Similarly, for $(j, l) \neq (j_{b-1}, l_{b-1})$ we have

$$P(\mathcal{A}_{2,b}(j, l)) \leq 2^{-N(I(U_1; Z_2 | U X_2) - 2\delta)}. \tag{19}$$

Thus, there is

$$\lim_{N \rightarrow \infty} P(\mathcal{A}) = 0 \tag{20}$$

whenever

$$\begin{aligned}
 R_{1,c} + R_{V_{1,c}} &\leq I(U_1; Z_2 | U X_2), \\
 R_{2,c} + R_{V_{2,c}} &\leq I(U_2; Z_1 | U X_1). \tag{21}
 \end{aligned}$$

By the classical results on source coding, there is

$$\lim_{N \rightarrow \infty} P(\mathcal{B}^c | \mathcal{A}^c) = 0 \tag{22}$$

whenever

$$\begin{aligned}
 R_{z_{1,c}} + R_{z_{1,p}} &\geq I(V_{1,c} V_{1,p}; U U_1 U_2 X_1 Z_1), \\
 R_{z_{2,c}} + R_{z_{2,p}} &\geq I(V_{2,c} V_{2,p}; U U_1 U_2 X_2 Z_2). \tag{23}
 \end{aligned}$$

We proceed to bound $P(\mathcal{G}_B | \mathcal{B}, \mathcal{A}^c)$. Event $\mathcal{G}_B$ conditioned on $\mathcal{B}, \mathcal{A}^c$ represents that the decoder obtains correct indices $\Delta_{1,c}, \Delta_{1,p}, \Delta_{2,c}, \Delta_{2,p}$ in block B . Since we have assumed that the decoder has at hand the correct bin indices $l_{B+1}, m_{B+1}, \beta_{1,B+1}, \beta_{2,B+1}$. Event $\mathcal{G}_B$ is the decoding error event in distributed Wyner-Ziv coding. By the law of large numbers, as $N \rightarrow \infty$, there is $P(\mathcal{G}^c(\Delta_{1,c,B+1}, \Delta_{1,p,B+1}, \Delta_{2,c,B+1}, \Delta_{2,p,B+1} | \mathcal{B}, \mathcal{A}^c) \rightarrow 0$. The probability of union in event $\mathcal{G}_B$ conditioned on $\mathcal{B}, \mathcal{A}^c$ is

$$\begin{aligned}
 &P\left(\bigcup_{\substack{(\Delta_{1,c}, \Delta_{1,p}, \Delta_{2,c}, \Delta_{2,p}) \neq \\ (\Delta_{1,c,B+1}, \Delta_{1,p,B+1}, \\ \Delta_{2,c,B+1}, \Delta_{2,p,B+1})}} \mathcal{G}(\Delta_{1,c}, \Delta_{1,p}, \Delta_{2,c}, \Delta_{2,p}) | \mathcal{B}, \mathcal{A}^c\right) \\
 &= P\left(\bigcup_{\substack{(\Delta_{1,c}, \Delta_{1,p}) \neq \\ (\Delta_{1,c,B+1}, \Delta_{1,p,B+1})}} \mathcal{G}(\Delta_{1,c}, \Delta_{1,p}, \Delta_{2,c,B+1}, \Delta_{2,p,B+1}) | \mathcal{B}, \mathcal{A}^c\right) \\
 &\quad + P\left(\bigcup_{\substack{(\Delta_{2,c}, \Delta_{2,p}) \neq \\ (\Delta_{2,c,B+1}, \Delta_{2,p,B+1})}} \mathcal{G}(\Delta_{1,c,B+1}, \Delta_{1,p,B+1}, \Delta_{2,c}, \Delta_{2,p}) | \mathcal{B}, \mathcal{A}^c\right) \\
 &\quad + P\left(\bigcup_{\substack{(\Delta_{1,c}, \Delta_{1,p}) \neq \\ (\Delta_{1,c,B+1}, \Delta_{1,p,B+1}), \\ \Delta_{2,c}, \Delta_{2,p} \neq \\ (\Delta_{2,c,B+1}, \Delta_{2,p,B+1})}} \mathcal{G}(\Delta_{1,c}, \Delta_{1,p}, \Delta_{2,c}, \Delta_{2,p}) | \mathcal{B}, \mathcal{A}^c\right), \tag{24}
 \end{aligned}$$

where $\Delta_{1,c} \in \alpha_{1,c}(l_{B+1}), \Delta_{1,p} \in \alpha_{1,p}(\beta_{1,B+1}), \Delta_{2,c} \in \alpha_{2,c}(m_{B+1}), \Delta_{2,p} \in \alpha_{2,p}(\beta_{2,B+1})$. By the Wyner-Ziv coding constraints [10], we conclude that

$$\lim_{N \rightarrow \infty} P(\mathcal{G}_B | \mathcal{B}, \mathcal{A}^c) = 0 \tag{25}$$

if

$$\begin{aligned}
 R_{z_{1,c}} - R_{V_{1,c}} + R_{z_{1,p}} - R_{V_{1,p}} &\leq I(V_{1,c} V_{1,p}; V_{2,c} V_{2,p} S_R Y), \\
 R_{z_{2,c}} - R_{V_{2,c}} + R_{z_{2,p}} - R_{V_{2,p}} &\leq I(V_{2,c} V_{2,p}; V_{1,c} V_{1,p} S_R Y), \\
 R_{z_{1,c}} - R_{V_{1,c}} + R_{z_{1,p}} - R_{V_{1,p}} + R_{z_{2,c}} - R_{V_{2,c}}
 \end{aligned}$$

$$\begin{aligned}
\mathcal{A}_{1,b}(k_{b-1}, m_{b-1}) &= \left(u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), \right. \\
&\quad \left. u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}), x_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}, r_{b-1}, \beta_{1,b-1}), z_1^N(b-1) \right) \in \mathcal{T}_\epsilon^N. \\
\mathcal{A}_{2,b}(j_{b-1}, l_{b-1}) &= \left(u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), \right. \\
&\quad \left. u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}), x_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}, s_{b-1}, \beta_{2,b-1}), z_2^N(b-1) \right) \in \mathcal{T}_\epsilon^N. \\
\mathcal{B}_{1,b} &= \left\{ \left(u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}), \right. \right. \\
&\quad \left. x_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}, r_{b-1}, \beta_{1,b-1}), z_1^N(b-1), \right. \\
&\quad \left. v_{1,c}^N(\Delta_{1,c,b}), v_{1,p}^N(\Delta_{1,p,b}) \right) \in \mathcal{T}_\epsilon^N, \text{ for some } \Delta_{1,c,b} \in [1 : 2^{NR_{z_{1,c}}}], \Delta_{1,p,b} \in [1 : 2^{NR_{z_{1,p}}}] \Big\} \\
\mathcal{B}_{2,b} &= \left\{ \left(u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}), \right. \right. \\
&\quad \left. x_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}, s_{b-1}, \beta_{2,b-1}), z_2^N(b-1), \right. \\
&\quad \left. v_{2,c}^N(\Delta_{2,c,b}), v_{2,p}^N(\Delta_{2,p,b}) \right) \in \mathcal{T}_\epsilon^N, \text{ for some } \Delta_{2,c,b} \in [1 : 2^{NR_{z_{2,c}}}], \Delta_{2,p,b} \in [1 : 2^{NR_{z_{2,p}}}] \Big\} \\
\mathcal{G}(\Delta_{1,c,b+1}, \Delta_{1,p,b+1}, \Delta_{2,c,b+1}, \Delta_{2,p,b+1}) &= \left\{ \left(v_{1,c}^N(\Delta_{1,c,b+1}), v_{1,p}^N(\Delta_{1,p,b+1}), v_{2,c}^N(\Delta_{2,c,b+1}), v_{2,p}^N(\Delta_{2,p,b+1}), \right. \right. \\
&\quad \left. s_R^N(b), y^N(b) \right) \in \mathcal{T}_\epsilon^N, \text{ for some } \Delta_{1,c,b+1} \in [1 : 2^{NR_{z_{1,c}}}], \Delta_{1,p,b+1} \in [1 : 2^{NR_{z_{1,p}}}], \\
&\quad \left. \Delta_{2,c,b+1} \in [1 : 2^{NR_{z_{2,c}}}], \Delta_{2,p,b+1} \in [1 : 2^{NR_{z_{2,p}}}] \right\} \\
\mathcal{H}(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r_b, \beta_{1,b}, s_b, \beta_{2,b}) &= \left\{ \left(u^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}), u_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b), \right. \right. \\
&\quad u_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b), x_1^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, j_b, l_b, r_b, \beta_{1,b}), x_2^N(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, k_b, m_b, s_b, \beta_{2,b}), \\
&\quad \left. v_{1,c}^N(\Delta_{1,c,b+1}), v_{1,p}^N(\Delta_{1,p,b+1}), v_{2,c}^N(\Delta_{2,c,b+1}), v_{2,p}^N(\Delta_{2,p,b+1}), s_R^N(b), y^N(b) \right) \in \mathcal{T}_\epsilon^N \Big\} \\
\mathcal{M}_{1,b}(\Delta_{2,c,b-1}) &= \left\{ u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), \right. \\
&\quad u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}), x_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}, r_{b-1}, \beta_{1,b-1}), \\
&\quad \left. z_1^N(b-1), v_{2,c}^N(\Delta_{2,c,b-1}) \right) \in \mathcal{T}_\epsilon^N, \text{ for some } \Delta_{2,c,b-1} \in [1 : 2^{NR_{z_{2,c}}}] \Big\} \\
\mathcal{M}_{2,b}(\Delta_{1,c,b-1}) &= \left\{ u^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}), u_1^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, j_{b-1}, l_{b-1}), \right. \\
&\quad u_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}), x_2^N(j_{b-2}, l_{b-2}, k_{b-2}, m_{b-2}, k_{b-1}, m_{b-1}, s_{b-1}, \beta_{2,b-1}), \\
&\quad \left. z_2^N(b-1), v_{1,c}^N(\Delta_{1,c,b-1}) \right) \in \mathcal{T}_\epsilon^N, \text{ for some } \Delta_{1,c,b-1} \in [1 : 2^{NR_{z_{1,c}}}] \Big\} \tag{15}
\end{aligned}$$

$$+R_{z_{2,p}} - R_{V_{2,p}} \leq I(V_{1,c}V_{1,p}V_{2,c}V_{2,p}; S_R Y). \quad (26) \quad \text{conditioned on } \mathcal{G}_B^c, \mathcal{B}, \mathcal{A}^c \text{ can be expressed as}$$

Next, we consider the probability $P(\mathcal{H}_B | \mathcal{G}_B^c, \mathcal{B}, \mathcal{A}^c)$. In block B , the decoder just needs to find j_{B-1} , l_{B-1} , k_{B-1} , m_{B-1} . Without loss of generality, we assume that $(l_B, m_B, \beta_{1,B}, \beta_{2,B}) = (1, 1, 1, 1)$. By the law of larger numbers, there is $P(\mathcal{H}^c(j_{B-1}, l_{B-1}, k_{B-1}, m_{B-1}, 1, 1, 1, 1 | \mathcal{G}_B^c, \mathcal{B}, \mathcal{A}^c))$ tends to zero as $N \rightarrow 0$. The probability of union in event $\mathcal{H}_B$

$$\begin{aligned}
&P\left(\bigcup_{(j,l,k,m) \neq (j_{B-1}, l_{B-1}, k_{B-1}, m_{B-1})} \mathcal{H}(j, l, k, m, 1, 1, 1, 1) \right) \\
&\leq 2^N (R_{1,c} + R_{2,c} + R_{V_{1,c}} + R_{V_{2,c}}) \\
&\quad 2^{-N(I(UU_1U_2X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y | S_R) - \delta)} \tag{27}
\end{aligned}$$

$$\begin{aligned}
\mathcal{A} &= \bigcup_{b=2}^B \left(\mathcal{A}_{1,b}^c(k_{b-1}, m_{b-1}) \bigcup (\cup_{(k,m) \neq (k_{b-1}, m_{b-1})} \mathcal{A}_{1,b}(k, m)) \right. \\
&\quad \left. \bigcup \mathcal{A}_{2,b}^c(j_{b-1}, l_{b-1}) \bigcup (\cup_{(j,l) \neq (j_{b-1}, l_{b-1})} \mathcal{A}_{2,b}(j, l)) \right) \\
\mathcal{B} &= \bigcup_{b=2}^{B+1} (\mathcal{B}_{1,b} \cap \mathcal{B}_{2,b}) \\
\mathcal{G}_b &= \mathcal{G}^c(\Delta_{1,c,b+1}, \Delta_{1,p,b+1}, \Delta_{2,c,b+1}, \Delta_{2,p,b+1}) \bigcup (\cup_{\substack{\Delta_{1,c} \in \alpha_{1,c}(l_{b+1}), \Delta_{1,p} \in \alpha_{1,p}(\beta_{1,b+1}) \\ \Delta_{2,c} \in \alpha_{2,c}(m_{b+1}), \Delta_{2,p} \in \alpha_{2,p}(\beta_{2,b+1}) \\ (\Delta_{1,c}, \Delta_{1,p}, \Delta_{2,c}, \Delta_{2,p}) \neq \\ (\Delta_{1,c,b+1}, \Delta_{1,p,b+1}, \Delta_{2,c,b+1}, \Delta_{2,p,b+1})}} \mathcal{G}(\Delta_{1,c}, \Delta_{1,p}, \Delta_{2,c}, \Delta_{2,p})) \\
\mathcal{H}_b &= \mathcal{H}^c(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r_b, \beta_{1,b}, s_b, \beta_{2,b}) \bigcup (\cup_{(j,l,k,m,r,\beta_1,s,\beta_2) \neq (j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r_b, \beta_{1,b}, s_b, \beta_{2,b})} \mathcal{H}(j, l, k, m, r, \beta_1, s, \beta_2)) \\
\mathcal{M} &= \bigcup_{b=2}^{B+1} \left(\mathcal{M}_{1,b}^c(\Delta_{2,c,b-1}) \bigcup (\cup_{\Delta_{2,c} \neq \Delta_{2,c,b-1}} \mathcal{M}_{1,b}(\Delta_{2,c})) \bigcup \mathcal{M}_{2,b}^c(\Delta_{1,c,b-1}) \bigcup (\cup_{\Delta_{1,c} \neq \Delta_{1,c,b-1}} \mathcal{M}_{2,b}(\Delta_{1,c})) \right) \quad (16)
\end{aligned}$$

Therefore, we conclude that

$$\lim_{N \rightarrow \infty} P(\mathcal{H}_B | \mathcal{G}_B^c, \mathcal{B}, \mathcal{A}^c) = 0 \quad (28)$$

if

$$R_{1,c} + R_{2,c} + R_{V_{1,c}} + R_{V_{2,c}} \leq I(UU_1 U_2 X_1 X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | S_R). \quad (29)$$

For block $b \in [2 : B-1]$, $P(\mathcal{G}_b | \mathcal{H}_{b+1}^c, \mathcal{B}, \mathcal{A}^c)$ is treated exactly as $P(\mathcal{G}_B | \mathcal{B}, \mathcal{A}^c)$. The analysis of term $P(\mathcal{H}_b | \mathcal{G}_b^c, \mathcal{H}_{b+1}^c, \mathcal{B}, \mathcal{A}^c)$ is similar to that of $P(\mathcal{H}_B | \mathcal{G}_B^c, \mathcal{B}, \mathcal{A}^c)$. The probability of the first term in $\mathcal{H}_b$ also tends to zero as $N \rightarrow \infty$. The probability of union in $\mathcal{H}_b$ can be expressed as

$$\begin{aligned}
&P\left(\bigcup_{\substack{(j,l,k,m,r,\beta_1,s,\beta_2) \neq \\ (j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r_b, \beta_{1,b}, s_b, \beta_{2,b})}} \mathcal{H}(j, l, k, m, r, \beta_1, s, \beta_2)\right) \\
&= P\left(\bigcup_{\substack{(j,l,k,m) \neq (j_{b-1}, l_{b-1}, \\ k_{b-1}, m_{b-1}), \quad r, \beta_1, s, \beta_2}} \mathcal{H}(j, l, k, m, r, \beta_1, s, \beta_2)\right) \\
&+ P\left(\bigcup_{(r,\beta_1) \neq (r_b, \beta_{1,b})} \mathcal{H}(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r, \beta_1, s_b, \beta_{2,b})\right) \text{ whenever} \\
&+ P\left(\bigcup_{(s,\beta_2) \neq (s_b, \beta_{2,b})} \mathcal{H}(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r_b, \beta_{1,b}, s, \beta_2)\right) \\
&+ P\left(\bigcup_{\substack{(r,\beta_1) \neq (r_b, \beta_{1,b}), \\ (s,\beta_2) \neq (s_b, \beta_{2,b})}} \mathcal{H}(j_{b-1}, l_{b-1}, k_{b-1}, m_{b-1}, r, \beta_1, s, \beta_2)\right).
\end{aligned}$$

Therefore, by the joint typicality lemma, we have

$$\lim_{N \rightarrow \infty} P(\mathcal{H}_b | \mathcal{G}_b^c, \mathcal{H}_{b+1}^c, \mathcal{B}, \mathcal{A}^c) = 0 \quad (30)$$

provided

$$\begin{aligned}
R_{1,p} + R_{V_{1,p}} &\leq I(X_1; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | UU_1 U_2 X_2 S_R), \\
R_{2,p} + R_{V_{2,p}} &\leq I(X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | UU_1 U_2 X_1 S_R), \\
R_{1,p} + R_{2,p} + R_{V_{1,p}} + R_{V_{2,p}} &\leq \\
&I(X_1 X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | UU_1 U_2 S_R),
\end{aligned}$$

$$R_{1,c} + R_{2,c} + R_{V_{1,c}} + R_{V_{2,c}} + R_{1,p} + R_{2,p} + R_{V_{1,p}} + R_{V_{2,p}} \leq I(UU_1 U_2 X_1 X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | S_R). \quad (31)$$

Next, we bound the probability of $P(\mathcal{G}_1 | \mathcal{H}_2^c, \mathcal{B}, \mathcal{A}^c)$ and $P(\mathcal{H}_1 | \mathcal{G}_1^c, \mathcal{H}_2^c, \mathcal{B}, \mathcal{A}^c)$. Term $P(\mathcal{G}_1 | \mathcal{H}_2^c, \mathcal{B}, \mathcal{A}^c)$ is treated exactly as $P(\mathcal{G}_B | \mathcal{B}, \mathcal{A}^c)$ and yields the same rate constraints, i.e., (26). The analysis of $P(\mathcal{H}_1 | \mathcal{G}_1^c, \mathcal{H}_2^c, \mathcal{B}, \mathcal{A}^c)$ is a special case of that of $P(\mathcal{H}_b | \mathcal{G}_b^c, \mathcal{H}_{b+1}^c, \mathcal{B}, \mathcal{A}^c)$, $b \in [2 : B-1]$, i.e., $j_0 = l_0 = k_0 = m_0 = 1$ and $l_1 = m_1 = \beta_{1,1} = \beta_{2,1} = 1$. By similar arguments

$$\lim_{N \rightarrow \infty} P(\mathcal{H}_1 | \mathcal{G}_1^c, \mathcal{H}_2^c, \mathcal{B}, \mathcal{A}^c) = 0 \quad (32)$$

provided

$$\begin{aligned}
R_{1,p} &\leq I(X_1; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | UU_1 U_2 X_2 S_R), \\
R_{2,p} &\leq I(X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | UU_1 U_2 X_1 S_R), \\
R_{1,p} + R_{2,p} &\leq I(X_1 X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | UU_1 U_2 S_R).
\end{aligned} \quad (33)$$

At last, we bound the probability of $P(\mathcal{M} | \mathcal{B}, \mathcal{A}^c)$. Using the standard arguments as in Wyner-Ziv coding [11], there is

$$\lim_{N \rightarrow \infty} P(\mathcal{M} | \mathcal{B}, \mathcal{A}^c) = 0 \quad (34)$$

$$\begin{aligned}
R_{z_{1,c}} - R_{V_{1,c}} &\leq I(V_{1,c}; UU_1 U_2 X_2 Z_2), \\
R_{z_{2,c}} - R_{V_{2,c}} &\leq I(V_{2,c}; UU_1 U_2 X_1 Z_1).
\end{aligned} \quad (35)$$

Therefore, to enable the transmission with arbitrarily small error probability, the rate constraints should be

$$\begin{aligned}
R_{1,c} + R_{V_{1,c}} &\leq I(U_1; Z_2 | U X_2), \\
R_{2,c} + R_{V_{2,c}} &\leq I(U_2; Z_1 | U X_1), \\
R_{z_{1,c}} + R_{z_{1,p}} &\geq I(V_{1,c} V_{1,p}; UU_1 U_2 X_1 Z_1), \\
R_{z_{2,c}} + R_{z_{2,p}} &\geq I(V_{2,c} V_{2,p}; UU_1 U_2 X_2 Z_2), \\
R_{z_{1,c}} - R_{V_{1,c}} + R_{z_{1,p}} - R_{V_{1,p}} &\leq I(V_{1,c} V_{1,p}; V_{2,c} V_{2,p} S_R Y), \\
R_{z_{2,c}} - R_{V_{2,c}} + R_{z_{2,p}} - R_{V_{2,p}} &\leq I(V_{2,c} V_{2,p}; V_{1,c} V_{1,p} S_R Y), \\
R_{z_{1,c}} - R_{V_{1,c}} + R_{z_{1,p}} - R_{V_{1,p}} + R_{z_{2,c}} - R_{V_{2,c}} \\
&\quad + R_{z_{2,p}} - R_{V_{2,p}} \leq I(V_{1,c} V_{1,p}; V_{2,c} V_{2,p} S_R Y) \\
&\quad + I(V_{2,c} V_{2,p}; V_{1,c} V_{1,p} S_R Y) - I(V_{1,c} V_{1,p}; V_{2,c} V_{2,p} | S_R Y),
\end{aligned}$$

$$\begin{aligned}
R_{1,p} + R_{V_{1,p}} &\leq I(X_1; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | U U_1 U_2 X_2 S_R), \\
R_{2,p} + R_{V_{2,p}} &\leq I(X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | U U_1 U_2 X_1 S_R), \\
R_{1,p} + R_{2,p} + R_{V_{1,p}} + R_{V_{2,p}} &\leq I(X_1 X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | U U_1 U_2 S_R), \\
R_{1,c} + R_{2,c} + R_{V_{1,c}} + R_{V_{2,c}} + R_{1,p} + R_{2,p} + R_{V_{1,p}} &+ R_{V_{2,p}} \leq I(U U_1 U_2 X_1 X_2; V_{1,c} V_{1,p} V_{2,c} V_{2,p} Y | S_R), \\
R_{z_{1,c}} - R_{V_{1,c}} &\leq I(V_{1,c}; U U_1 U_2 X_2 Z_2), \\
R_{z_{2,c}} - R_{V_{2,c}} &\leq I(V_{2,c}; U U_1 U_2 X_1 Z_1). \tag{36}
\end{aligned}$$

Defining $R_1 = R_{1,p} + R_{1,c}$, $R_2 = R_{2,p} + R_{2,c}$, and by Fourier-Motzkin elimination approach, one can eliminate auxilarly RVs $R_{V_{1,c}}$, $R_{V_{1,p}}$, $R_{V_{2,c}}$, $R_{V_{2,p}}$, $R_{z_{1,c}}$, $R_{z_{1,p}}$, $R_{z_{2,c}}$, $R_{z_{2,p}}$, and obtain the achievable rate region as stated in the main results.

APPENDIX B ANALYSIS OF EXPECTED DISTORTION

Define $w_k = \{w_k^{(b)}\}_{b=1}^{B-1}$, $k \in \{1, 2\}$ with $|\mathcal{W}_k| = 2^{N(B-1)(R_{k,p}+R_{k,c})}$ where $w_k^{(b)} = \{w_{k,p}^{(b)}, w_{k,c}^{(b)}\}$. For any given message pair (w_1, w_2) , the expected distortion of transmission 1 is

$$\begin{aligned}
\limsup_{n \rightarrow \infty} d_1^{(n)}(w_1, w_2) &= \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n d_1(S_{T_1,i}, \hat{S}_{T_1,i}) \right] \\
&\stackrel{(a)}{\leq} \limsup_{n \rightarrow \infty} \left(P_e^{(n)} d_{\max} \right. \\
&\quad \left. + (1 - P_e^{(n)})(1 + \epsilon) \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n d_1(S_{T_1,i}, \hat{S}_{T_1,i}) \right] \right) \\
&\stackrel{(b)}{\leq} \limsup_{n \rightarrow \infty} \left(P_e^{(n)} d_{\max} + (1 - P_e^{(n)})(1 + \epsilon) D_1 \right) \\
&\stackrel{(c)}{=} D_1 \tag{37}
\end{aligned}$$

where (a) follows by applying the upper bound of the distortion function to the decoding error event and the typical average lemma [12] to the successful decoding event; (b) follows from the random codebook generation and state estimating function $\hat{s}_{T_1}(x_1, u_2, z_1)$ that achieves D_1 ; (c) follows because $P_e^{(n)}$ tends to zeros as $n \rightarrow \infty$ if the rate constraints (36) holds. Since the uniformly distributed messages are considered, it is easy to show that

$$\limsup_{n \rightarrow \infty} d_1^{(n)} \leq D_1. \tag{38}$$

The similar analysis can be conducted for transmitter 2 to verify that

$$\limsup_{n \rightarrow \infty} d_2^{(n)} \leq D_2. \tag{39}$$

APPENDIX C DISCUSSION OF THE TERMINATION BLOCKS

In the last $\tilde{B}$ termination blocks, we need to guarantee that the decoder can successfully find the indices $l_B, m_B, \beta_{1,B}, \beta_{2,B}, l_{B+1}, m_{B+1}, \beta_{1,B+1}, \beta_{2,B+1}$, i.e., the bin number of compressed sequences. Since the total bits of the compressed sequences are finite, we only need to prove that both transmitter 1 and 2 have the positive transmission rate.

Then, the number of termination blocks is finite and can be omitted when $B \rightarrow \infty$.

We first show that

$$R_1 + R_2 \leq I(X_1 X_2; Y | S_R), \tag{40}$$

which is given as follows.

$$\begin{aligned}
n(R_1 + R_2) &\stackrel{(a)}{\leq} I(W_1, W_2; Y^n | S_R^n) + n\epsilon_n \\
&= H(Y^n | S_R^n) - H(Y^n | S_R^n, W_1, W_2) + n\epsilon_n \\
&= \sum_{i=1}^n H(Y_i | Y^{i-1}, S_R^n) - H(Y_i | Y^{i-1}, S_R^n, W_1, W_2) + n\epsilon_n \\
&\stackrel{(b)}{\leq} \sum_{i=1}^n H(Y_i | S_{R,i}) - H(Y_i | Y^{i-1}, S_R^n, W_1, W_2, X_{1,i}, X_{2,i}) + n\epsilon_n \\
&\stackrel{(c)}{=} \sum_{i=1}^n H(Y_i | S_{R,i}) - H(Y_i | S_{R,i}, X_{1,i}, X_{2,i}) + n\epsilon_n \\
&= \sum_{i=1}^n I(X_{1,i}, X_{2,i}; Y_i | S_{R,i}) + n\epsilon_n, \tag{41}
\end{aligned}$$

where (a) follows from Fano's inequality; (b) follows because conditioning reduces entropy; (c) follows because $(W_1, W_2, Y^{i-1}, \{S_{R,l}\}_{l \neq i}) - (X_{1,i}, X_{2,i}, S_{R,i}) - Y_i$ forms a Markov chain.

Based on (40), we known that if $I(X_1 X_2; Y | S_R) = 0$, there must be $R_1 = R_2 = 0$ and hence achievable. It thus remains to consider when

$$I(X_1 X_2; Y | S_R) > 0. \tag{42}$$

Since

$$\begin{aligned}
I(X_1 X_2; Y | S_R) &= I(X_1; Y | S_R) + I(X_2; Y | X_1 S_R), \tag{43} \\
I(X_1; Y | S_R) \text{ and } I(X_2; Y | X_1 S_R) &\text{ cannot be both zero. This reveals that at least one transmitter can achieve a positive transmission rate. Without loss of generality, we assume that transmitter 1 can achieve a positive transmission rate } r_1. \text{ With } r_1 > 0, \text{ the transmitter 1 can successfully transmit the indices } l_B, \beta_{1,B}, l_{B+1}, \beta_{1,B+1} \text{ within finite blocks.}
\end{aligned}$$

Next, we notice that due to the message cooperation, transmitter 1 can get partial information of the codewords of transmitter 2, i.e., U_2 . If $U_2 \neq U$, i.e., transmitter 1 can get the information of common messages of transmitter 2, the remaining indices $m_B, \beta_{2,B}, m_{B+1}, \beta_{2,B+1}$ associated to transmitter 2 can also be successfully decoded by the decoder. The operation is as follows. The transmitter 2 sends a lossless description of remaining indices to transmitter 1. Next, transmitter 1 sends a lossless description of remaining indices to the decoder. Such operations can be done in finite blocks. Therefore, if $U_2 \neq U$, all remaining indices can be successfully decoded by the decoder.

Now, it only remains to prove that if $U_2 = U$, transmitter 2 can still successfully transmit the remaining indices. Since transmitter 1 has a positive rate r_1 , it can send a lossless description of $X_1 U_2 Z_1$ within finite blocks. With the help of $X_1 U_2 Z_1$, the transmission rate of transmitter 2 is $I(X_2; Y X_1 U_2 Z_1 | S_R)$. If $I(X_2; Y X_1 U_2 Z_1 | S_R) > 0$, the decoder can successfully decode the information transmitted by transmitter 2 thus obtain the remaining indices. Thus, we only need to consider the case $I(X_2; Y X_1 U_2 Z_1 | S_R) = 0$.

In this case, we prove that if

$$I(X_2; Y X_1 U_2 Z_1 | S_R) = 0, \quad (44)$$

there must be

$$\begin{aligned} R_2 &\leq I(U_2; Z_1 | U X_1) + I(X_2; Y | U U_2 X_1 V_1 V_2 S_R) \\ &\quad - I(V_2; X_2 U_1 Z_2 | V_1 S_R Y), \\ &\leq 0, \end{aligned} \quad (45)$$

which means that if $I(X_2; Y X_1 U_2 Z_1 | S_R) = 0$, the rate pair $(R_1, R_2 = 0)$ under the sensing constraints is achievable by considering the MAC as a single-user channel from X_1 to Y and by letting transmitter 2 sends nothing or sends a deterministic sequence. When $U_2 = U$, there is $I(U_2; Z_1 | U X_1) = 0$. Thus, there is

$$\begin{aligned} R_2 &\leq I(X_2; Y | U X_1 V_1 V_2 S_R) - I(V_2; X_2 U_1 Z_2 | V_1 S_R Y), \\ &\stackrel{(a)}{\leq} I(X_2; Y X_1 U Z_1 | U X_1 V_1 V_2 S_R) \\ &\quad - I(V_2; X_2 U_1 Z_2 | X_1 U Z_1 V_1 S_R Y), \end{aligned} \quad (46)$$

where (a) follows because introducing $X_1 U Z_1$ in the first term does not decrease the mutual information, and $V_2 - X_2 U_1 Z_2 - X_1 U Z_1$ forms a Markov chain. Since

$$\begin{aligned} &I(X_2; Y X_1 U Z_1 | U X_1 V_1 V_2 S_R) \\ &= H(X_2 | U X_1 V_1 V_2 S_R) - H(X_2 | Y X_1 U Z_1 V_1 V_2 S_R) \\ &\stackrel{(a)}{=} H(X_2 | U X_1 V_2 S_R) - H(X_2 | Y X_1 U Z_1 V_2 S_R) \\ &= I(X_2; Y Z_1 | U X_1 V_2 S_R) \\ &\leq I(X_2; Y U X_1 Z_1 | U X_1 V_2 S_R), \end{aligned} \quad (47)$$

where (a) follows because in the first term, given $U_2 = U$, X_2 and $X_1 V_1 V_2 S_R$ are independent, and in the second term, $X_2 - X_1 U Z_1 - V_1$ forms a Markov chain, and

$$\begin{aligned} &I(V_2; X_2 U_1 Z_2 | X_1 U Z_1 V_1 S_R Y) \\ &= H(V_2 | X_1 U Z_1 V_1 S_R Y) - H(V_2 | X_2 U_1 Z_2) \\ &\stackrel{(b)}{=} H(V_2 | X_1 U Z_1 S_R Y) - H(V_2 | X_2 U_1 Z_2) \\ &= I(V_2; X_2 U_1 Z_2 | X_1 U Z_1 S_R Y), \end{aligned} \quad (48)$$

where (b) follows because $V_2 - X_1 U Z_1 - V_1$ forms a Markov chain, there is

$$\begin{aligned} R_2 &\leq I(X_2; Y X_1 U Z_1 | U X_1 V_2 S_R) \\ &\quad - I(V_2; X_2 U_1 Z_2 | X_1 U Z_1 S_R Y), \\ &\leq I(X_2; Y X_1 U Z_1 V_2 | S_R) \\ &\quad - I(V_2; X_2 U_1 Z_2 | X_1 U Z_1 S_R Y), \\ &= I(X_2; Y X_1 U Z_1 | S_R) + I(X_2; V_2 | Y X_1 U Z_1 S_R) \\ &\quad - I(V_2; X_2 U_1 Z_2 | X_1 U Z_1 S_R Y), \\ &\stackrel{(c)}{\leq} I(X_2; V_2 | Y X_1 U Z_1 S_R) \\ &\quad - I(V_2; X_2 U_1 Z_2 | X_1 U Z_1 S_R Y), \\ &= -I(V_2; U_1 Z_2 | X_2 Y X_1 U Z_1 S_R) \\ &\leq 0, \end{aligned} \quad (49)$$

where (c) follows because the assumption (44) holds. To this end, if $I(X_2; Y X_1 U_2 Z_1 | S_R) = 0$, the rate pair $(R_1, R_2 = 0)$ under the sensing constraints is achievable, which completes the proof.

APPENDIX D

CALCULATION OF TERMS IN ACHIEVABLE RATE RESULT OF OUR SCHEME

Calculation of term $I(U_k; Z_k | U X_k)$: We first calculate term $I(U_1; Z_2 | U X_2)$. There is

$$\begin{aligned} &I(U_1; Z_2 | U X_2) \\ &= H(Z_2 | U X_2) - H(Z_2 | U U_1 X_2) \\ &= H(\Sigma_1 \oplus \Theta_1 \oplus S_{21} \oplus B) - H(\Theta_1 \oplus S_{21} \oplus B) \\ &= H_2(p_{\Sigma_1} * p_{\Theta_1} * p_{S_{21}} * p_B) - H_2(p_{\Theta_1} * p_{S_{21}} * p_B). \end{aligned} \quad (50)$$

Since $\Sigma_2 = 0$, i.e., $U_2 = U$, it is obvious that

$$I(U_2; Z_1 | U X_1) = 0. \quad (51)$$

Calculation of term $I(V_{k,c}, V_{k,p}; U U_1 U_2 X_k Z_k)$: It is quite obvious that

$$I(V_{2,c}, V_{2,p}; U U_1 U_2 X_2 Z_2) = 0. \quad (52)$$

Then, we focus on the calculation of $I(V_{1,c}, V_{1,p}; U U_1 U_2 X_1 Z_1)$. There is

$$\begin{aligned} &I(V_{1,c}, V_{1,p}; U U_1 U_2 X_1 Z_1) \\ &= H(V_{1,c}, V_{1,p}) - H(V_{1,c}, V_{1,p} | X_1 Z_1) \\ &= H(V_{1,c}, V_{1,p}) - H(V_{1,c}, V_{1,p} | S_{\text{comb}}) \\ &= H(S_{\text{comb}}) - H(S_{\text{comb}} | V_{1,c}, V_{1,p}) \\ &= H_2(p_{S_{21}} * p_{S_{22}}) - H_2(p_{N_{1,0}} * p_{N_{2,0}}), \end{aligned} \quad (53)$$

where $S_{\text{comb}} \triangleq S_{21} \oplus S_{22}$.

Calculation of term $I(V_{k,c}, V_{k,p}; V_{k,c}, V_{k,p} S_R Y)$: It is obvious that

$$I(V_{2,c}, V_{2,p}; V_{1,c}, V_{1,p} S_R Y) = 0. \quad (54)$$

We proceed to calculate $I(V_{1,c}, V_{1,p}; V_{2,c}, V_{2,p} S_R Y)$. There is

$$\begin{aligned} &I(V_{1,c}, V_{1,p}; V_{2,c}, V_{2,p} S_R Y) = H(Y) - H(Y | V_{1,c}, V_{1,p}) \\ &= H(U \oplus \Sigma_1 \oplus \Theta_1 \oplus S_1, U \oplus \Theta_2 \oplus S_{21} \oplus S_{22}) \\ &\quad - H(U \oplus \Sigma_1 \oplus \Theta_1 \oplus S_1, U \oplus \Theta_2 \oplus N_{1,0} \oplus N_{2,0}) \\ &= H_4(b_1, b_2, b_3, b_4) - H_4(c_1, c_2, c_3, c_4), \end{aligned} \quad (55)$$

where

$$\begin{aligned} b_1 &= p_U(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\ &\quad + (1 - p_U)(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}), \\ b_2 &= p_U(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\ &\quad + (1 - p_U)(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}), \\ b_3 &= p_U(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\ &\quad + (1 - p_U)(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}), \\ b_4 &= p_U(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\ &\quad + (1 - p_U)(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}), \end{aligned} \quad (56)$$

and

$$\begin{aligned} c_1 &= p_U(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}) \\ &\quad + (1 - p_U)(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}), \\ c_2 &= p_U(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}) \\ &\quad + (1 - p_U)(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}), \\ c_3 &= p_U(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}) \\ &\quad + (1 - p_U)(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}), \\ c_4 &= p_U(1 - p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}) \end{aligned}$$

$$\begin{aligned}
q_1 &= \frac{(p_{\tilde{S}_{21}} p_{\tilde{S}_{22}} (1 - p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}})(1 - p_{\tilde{S}_{22}})(p_{N_{1,1}} * p_{N_{1,0}}))(p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}} p_{\tilde{S}_{22}} (p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}})(1 - p_{\tilde{S}_{22}})(1 - p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}})}, \\
q_2 &= \frac{(p_{\tilde{S}_{21}} (1 - p_{\tilde{S}_{22}})(1 - p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}}) p_{\tilde{S}_{22}} (p_{N_{1,1}} * p_{N_{1,0}}))(p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}} (1 - p_{\tilde{S}_{22}})(p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}}) p_{\tilde{S}_{22}} (1 - p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}})}, \\
q_3 &= \frac{(p_{\tilde{S}_{21}} p_{\tilde{S}_{22}} (1 - p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}})(1 - p_{\tilde{S}_{22}})(p_{N_{1,1}} * p_{N_{1,0}}))(1 - p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}} p_{\tilde{S}_{22}} (1 - p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}})(1 - p_{\tilde{S}_{22}})(p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}})}, \\
q_4 &= \frac{(p_{\tilde{S}_{21}} (1 - p_{\tilde{S}_{22}})(1 - p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}}) p_{\tilde{S}_{22}} (p_{N_{1,1}} * p_{N_{1,0}}))(1 - p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}} (1 - p_{\tilde{S}_{22}})(1 - p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}}) + (1 - p_{\tilde{S}_{21}}) p_{\tilde{S}_{22}} (p_{\Theta_1} * p_B * p_{N_{1,1}} * p_{N_{1,0}})}. \quad (67)
\end{aligned}$$

$$+ (1 - p_U)(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1 - p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}). \quad (57)$$

Calculation of term $I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}|S_R Y)$: It is obvious that

$$I(V_{1,c}V_{1,p}; V_{2,c}V_{2,p}|S_R Y) = 0. \quad (58)$$

Calculation of term $I(X_k; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_kS_R)$: We first calculate $I(X_1; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_1S_R)$. There is

$$\begin{aligned}
&I(X_1; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_1S_R) \\
&= I(X_1; Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}X_1S_R) \\
&= H(Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}X_1S_R) \\
&\quad - H(Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}X_1X_2S_R) \\
&= H(\Theta_1 \oplus S_1) - H(S_1) \\
&= H_2(p_{\Theta_1} * p_{S_1}) - H_2(p_{S_1}). \quad (59)
\end{aligned}$$

Similarly, for term $I(X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_1S_R)$, we have

$$\begin{aligned}
&I(X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2X_1S_R) \\
&= I(X_2; Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}X_1S_R) \\
&= H(Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}X_1S_R) \\
&\quad - H(Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}X_1X_2S_R) \\
&= H(\Theta_2 \oplus N_{1,0} \oplus N_{2,0}) - H(N_{1,0} \oplus N_{2,0}) \\
&= H_2(p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}) - H_2(p_{N_{1,0}} * p_{N_{2,0}}). \quad (60)
\end{aligned}$$

Calculation of term $I(X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2S_R)$: The calculation is given as follows.

$$\begin{aligned}
&I(X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|UU_1U_2S_R) \\
&= I(X_1X_2; Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}S_R) \\
&= H(Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}S_R) \\
&\quad - H(Y|UU_1U_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}X_1X_2S_R) \\
&= H(\Theta_1 \oplus S_1) + H(\Theta_2 \oplus N_{1,0} \oplus N_{2,0}) \\
&\quad - H(S_1) - H(N_{1,0} \oplus N_{2,0}) \\
&= H_2(p_{\Theta_1} * p_{S_1}) + H_2(p_{\Theta_2} * p_{N_{1,0}} * p_{N_{2,0}}) \\
&\quad - H_2(p_{S_1}) - H_2(p_{N_{1,0}} * p_{N_{2,0}}). \quad (61)
\end{aligned}$$

Calculation of term $I(UU_1U_2X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|S_R)$: There is

$$\begin{aligned}
&I(UU_1U_2X_1X_2; V_{1,c}V_{1,p}V_{2,c}V_{2,p}Y|S_R) \\
&= I(UU_1U_2X_1X_2; Y|V_{1,c}V_{1,p}V_{2,c}V_{2,p}S_R) \\
&= H(Y|V_{1,c}V_{1,p}V_{2,c}V_{2,p}S_R) \\
&\quad - H(Y|UU_1U_2X_1X_2V_{1,c}V_{1,p}V_{2,c}V_{2,p}S_R)
\end{aligned}$$

$$\begin{aligned}
&= H(U \oplus \Sigma_1 \oplus \Theta_1 \oplus S_1, U \oplus \Theta_2 \oplus N_{1,0} \oplus N_{2,0}) \\
&\quad - H(S_1) - H(N_{1,0} \oplus N_{2,0}) \\
&= H_4(c_1, c_2, c_3, c_4) - H_2(p_{S_1}) - H_2(p_{N_{1,0}} * p_{N_{2,0}}). \quad (62)
\end{aligned}$$

APPENDIX E

CALCULATION OF TERMS IN DISTORTION RESULT OF OUR ACHIEVABLE SCHEME

Since the sensing state of transmitter 2

$$S_{T_2} = S_{21}, \quad (63)$$

the state estimator of transmitter 2 is

$$\begin{aligned}
&\hat{S}_{21}(X_2, Z_2, U_1, V_{1,c}) = \arg \min_{s'_{21} \in \{0,1\}} \\
&\sum_{s'_{21}} P_{S_{21}|X_2Z_2U_1V_{1,c}}(s_{21}|x_2z_2u_1v_{1,c})d(s_{21}, s'_{21}). \quad (64)
\end{aligned}$$

Let $Z_2'' = \Theta_1 \oplus B \oplus \tilde{S}_{21} \oplus N_{1,1} \oplus N_{1,0}$. It is easy to obtain that

$$\begin{aligned}
&\hat{S}_{21}(X_2, Z_2, U_1, V_{1,c}) = \hat{S}_{21}(Z_2'', V_1) \triangleq \arg \min_{s'_{21} \in \{0,1\}} \\
&\sum_{s'_{21}} P_{S_{21}|Z_2''V_1}(s_{21}|z_2''v_1)d(s_{21}, s'_{21}). \quad (65)
\end{aligned}$$

Then, it is quite obvious that

$$\begin{aligned}
&\hat{S}_{21}(Z_2'' = 0, V_1 = 0) = \mathbb{1}\{q_1 \geq 0.5\}, \\
&\hat{S}_{21}(Z_2'' = 0, V_1 = 1) = \mathbb{1}\{q_2 \geq 0.5\}, \\
&\hat{S}_{21}(Z_2'' = 1, V_1 = 0) = \mathbb{1}\{q_3 \geq 0.5\}, \\
&\hat{S}_{21}(Z_2'' = 1, V_1 = 1) = \mathbb{1}\{q_4 \geq 0.5\}, \quad (66)
\end{aligned}$$

where q_1, q_2, q_3, q_4 are given in (67). Then, the expected distortion D_2 over all Z_2', V_1 based on the state estimator in (66) is given by

$$\begin{aligned}
D_2 &= P_{Z_2''V_1}(0, 0) \min\{q_1, 1 - q_1\} \\
&\quad + P_{Z_2''V_1}(0, 1) \min\{q_2, 1 - q_2\} \\
&\quad + P_{Z_2''V_1}(1, 0) \min\{q_3, 1 - q_3\} \\
&\quad + P_{Z_2''V_1}(1, 1) \min\{q_4, 1 - q_4\}. \quad (68)
\end{aligned}$$

APPENDIX F

CALCULATION OF TERMS IN ACHIEVABLE RATE RESULT OF SCHEME PROPOSED IN [9]

Calculation of term $I(V_k; X_kZ_k|UU_1U_2)$: It is quite obvious that

$$I(V_2; X_2Z_2|UU_1U_2) = 0. \quad (69)$$

Thus, we focus on the calculation of $I(V_1; X_1 Z_1 | UU_1 U_2)$. There is

$$I(V_1; X_1 Z_1 | UU_1 U_2) = H(V_1 | UU_1 U_2) - H(V_1 | UU_1 U_2 X_1 Z_1) = H(V_1) - H(V_1 | X_1 Z_1). \quad (70)$$

Let $S_{\text{comb}} = S_{21} \oplus S_{22}$. There is

$$H(V_1 | X_1 Z_1) = H(V_1 | S_{\text{comb}}) = H(S_{\text{comb}} | V_1) + H(V_1) - H(S_{\text{comb}}). \quad (71)$$

Thus, we have

$$I(V_1; X_1 Z_1 | UU_1 U_2) = H(S_{\text{comb}}) - H(S_{\text{comb}} | V_1) = H_2(p_{S_{21}} * p_{S_{22}}) - H_2(p_{N_1} * p_{N_2}). \quad (72)$$

Calculation of term $I(U_k; X_{\bar{k}} Z_{\bar{k}} | UU_{\bar{k}})$: We first calculate term $I(U_1; X_2 Z_2 | UU_2)$. The result is provided as follows.

$$\begin{aligned} I(U_1; X_2 Z_2 | UU_2) &= H(X_2 Z_2 | U) - H(X_2 Z_2 | UU_1) \\ &= H(Z_2 | U X_2) - H(Z_2 | UU_1 X_2) \\ &= H(\Sigma_1 \oplus \Theta_1 \oplus S_{21} \oplus B) - H(\Theta_1 \oplus S_{21} \oplus B) \\ &= H_2(p_{\Sigma_1} * p_{\Theta_1} * p_{S_{21}} * p_B) - H_2(p_{\Theta_1} * p_{S_{21}} * p_B). \end{aligned} \quad (73)$$

Since $\Sigma_2 = 0$, i.e., $U_2 = U$, it is obvious that

$$I(U_2; X_1 Z_1 | UU_1) = 0. \quad (74)$$

Calculation of term $I(V_k; X_{\bar{k}} Z_{\bar{k}} | UU_1 U_2)$: It is obvious that

$$I(V_2; X_1 Z_1 | UU_1 U_2) = 0. \quad (75)$$

For term $I(V_1; X_2 Z_2 | UU_1 U_2)$, there is

$$\begin{aligned} I(V_1; X_2 Z_2 | UU_1 U_2) &= H(X_2 Z_2 | UU_1 U_2) - H(X_2 Z_2 | UU_1 U_2 V_1) \\ &= H(X_2 | UU_1 U_2) + H(Z_2 | UU_1 U_2 X_2) - H(X_2 | UU_1 U_2 V_1) \\ &\quad - H(Z_2 | UU_1 U_2 V_1 X_2) \\ &= H(Z_2 | UU_1 U_2 X_2) - H(Z_2 | UU_1 U_2 V_1 X_2) \\ &= H(\Theta_1 \oplus S_{21} \oplus B) - H(\Theta_1 \oplus S_{21} \oplus B | V_1) \\ &= H_2(p_{\Theta_1} * p_{S_{21}} * p_B) - H(\Theta_1 \oplus S_{21} \oplus B | V_1). \end{aligned} \quad (76)$$

Since

$$\begin{aligned} H(\Theta_1 \oplus S_{21} \oplus B | V_1) &= H(\Theta_1 \oplus B \oplus N_1 \oplus \tilde{S}_{21} | V_1) \\ &= p(V_1 = 1)H(\Theta_1 \oplus B \oplus N_1 \oplus \tilde{S}_1 | V_1 = 1) \\ &\quad + p(V_1 = 0)H(\Theta_1 \oplus B \oplus N_1 \oplus \tilde{S}_1 | V_1 = 0) \\ &= H_4(a_1, a_2, a_3, a_4) - H_2(p_{\tilde{S}_1} * p_{\tilde{S}_2}), \end{aligned} \quad (77)$$

where

$$\begin{aligned} a_1 &= p_{\tilde{S}_{21}}(1 - p_{\tilde{S}_{22}})(1 - p_{\Theta_1} * p_B * p_{N_1}) \\ &\quad + (1 - p_{\tilde{S}_{21}})p_{\tilde{S}_{22}}(p_{\Theta_1} * p_B * p_{N_1}), \\ a_2 &= p_{\tilde{S}_{21}}(1 - p_{\tilde{S}_{22}})(p_{\Theta_1} * p_B * p_{N_1}) \\ &\quad + (1 - p_{\tilde{S}_{21}})p_{\tilde{S}_{22}}(1 - p_{\Theta_1} * p_B * p_{N_1}), \\ a_3 &= p_{\tilde{S}_{21}}p_{\tilde{S}_{22}}(1 - p_{\Theta_1} * p_B * p_{N_1}) \\ &\quad + (1 - p_{\tilde{S}_{21}})(1 - p_{\tilde{S}_{22}})(p_{\Theta_1} * p_B * p_{N_1}), \\ a_4 &= p_{\tilde{S}_{21}}p_{\tilde{S}_{22}}(p_{\Theta_1} * p_B * p_{N_1}) \\ &\quad + (1 - p_{\tilde{S}_{21}})(1 - p_{\tilde{S}_{22}})(1 - p_{\Theta_1} * p_B * p_{N_1}), \end{aligned} \quad (78)$$

we conclude that

$$I(V_1; X_2 Z_2 | UU_1 U_2) = H_2(p_{\Theta_1} * p_{S_{21}} * p_B) + H_2(p_{\tilde{S}_{21}} * p_{\tilde{S}_{22}}) - H_4(a_1, a_2, a_3, a_4). \quad (79)$$

Calculation of term $I(X_k; YV_1 V_2 | UU_1 U_2 X_{\bar{k}})$: We first cal-

culate term $I(X_1; YV_1 V_2 | UU_1 U_2 X_2)$. There is

$$\begin{aligned} I(X_1; YV_1 V_2 | UU_1 U_2 X_2) &= I(X_1; Y | UU_1 U_2 V_1 V_2 X_2) \\ &= H(Y | UU_1 U_2 V_1 V_2 X_2) - H(Y | UU_1 U_2 V_1 V_2 X_1 X_2) \\ &= H(\Theta_1 \oplus S_1) + H(N_1 \oplus N_2) - H(S_1) - H(N_1 \oplus N_2) \\ &= H_2(p_{\Theta_1} * p_{S_1}) - H_2(p_{S_1}). \end{aligned} \quad (80)$$

Similarly, for term $I(X_2; YV_1 V_2 | UU_1 U_2 X_1)$, we have

$$\begin{aligned} I(X_2; YV_1 V_2 | UU_1 U_2 X_1) &= I(X_2; Y | UU_1 U_2 V_1 V_2 X_1) \\ &= H(Y | UU_1 U_2 V_1 V_2 X_1) - H(Y | UU_1 U_2 V_1 V_2 X_1 X_2) \\ &= H(S_1) + H(\Theta_2 \oplus N_1 \oplus N_2) - H(S_1) - H(N_1 \oplus N_2) \\ &= H_2(p_{\Theta_2} * p_{N_1} * p_{N_2}) - H_2(p_{N_1} * p_{N_2}). \end{aligned} \quad (81)$$

Calculation of term $I(X_k; Y | U X_{\bar{k}})$: The calculation of term $I(X_1; Y | U X_2)$ is as follows.

$$\begin{aligned} I(X_1; Y | U X_2) &= I(X_1; Y_1 | U X_2) \\ &= H(Y_1 | U X_2) - H(Y_1 | X_1) \\ &= H(\Sigma_1 \oplus \Theta_1 \oplus S_1) - H(S_1) \\ &= H_2(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1}) - H_2(p_{S_1}). \end{aligned} \quad (82)$$

Similarly, we have

$$\begin{aligned} I(X_2; Y | U X_1) &= I(X_2; Y_2 | U X_1) \\ &= H(\Theta_2 \oplus S_{21} \oplus S_{22}) - H(S_{21} \oplus S_{22}) \\ &= H_2(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) - H_2(p_{S_{21}} * p_{S_{22}}). \end{aligned} \quad (83)$$

Calculation of term $I(V_1; X_1 X_2 Y | UU_1 U_2)$ and $I(V_2; X_1 X_2 YV_1 | UU_1 U_2)$: It is obvious that

$$I(V_2; X_1 X_2 YV_1 | UU_1 U_2) = 0. \quad (84)$$

The calculation of $I(V_1; X_1 X_2 Y | UU_1 U_2)$ is as follows.

$$\begin{aligned} I(V_1; X_1 X_2 Y | UU_1 U_2) &= H(X_1 X_2 Y | UU_1 U_2) - H(X_1 X_2 Y | UU_1 U_2 V_1) \\ &= H(Y | X_1 X_2) - H(Y | X_1 X_2 V_1) \\ &= H(S_1) + H(S_{\text{comb}}) - H(S_1) - H(N_1 \oplus N_2) \\ &= H_2(p_{S_{21}} * p_{S_{22}}) - H_2(p_{N_1} * p_{N_2}). \end{aligned} \quad (85)$$

Calculation of term $I(X_1 X_2; Y | UU_k)$: We first calculate term $I(X_1 X_2; Y | UU_1)$. There is

$$\begin{aligned} I(X_1 X_2; Y | UU_1) &= H(Y | UU_1) - H(Y | X_1 X_2) \\ &= H(Y_1 | UU_1) + H(Y_2 | Y_1 UU_1) - H(S_1) - H(S_{\text{comb}}) \\ &= H(\Theta_1 \oplus S_1) + H(\Theta_2 \oplus S_{21} \oplus S_{22}) - H(S_1) - H(S_{\text{comb}}) \\ &= H_2(p_{\Theta_1} * p_{S_1}) + H_2(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\ &\quad - H_2(p_{S_1}) - H_2(p_{S_{21}} * p_{S_{22}}). \end{aligned} \quad (86)$$

Similarly, there is

$$\begin{aligned} I(X_1 X_2; Y | UU_2) &= I(X_1 X_2; Y | U) \\ &= H(Y | U) - H(S_1) - H(S_{\text{comb}}) \\ &= H(\Sigma_1 \oplus \Theta_1 \oplus S_1) + H(\Theta_2 \oplus S_{21} \oplus S_{22}) \\ &\quad - H(S_1) - H(S_{\text{comb}}) \\ &= H_2(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1}) + H_2(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\ &\quad - H_2(p_{S_1}) - H_2(p_{S_{21}} * p_{S_{22}}). \end{aligned} \quad (87)$$

Calculation of term $I(X_1 X_2; YV_1 V_2 | UU_1 U_2)$: We have

$$\begin{aligned} I(X_1 X_2; YV_1 V_2 | UU_1 U_2) &= I(X_1 X_2; Y | UU_1 U_2 V_1 V_2) \\ &= H(Y | UU_1 U_2 V_1 V_2) - H(Y | X_1 X_2 UU_1 U_2 V_1 V_2) \\ &= H(\Theta_1 \oplus S_1) + H(\Theta_2 \oplus N_1 \oplus N_2) \end{aligned}$$

$$\begin{aligned}
p_1 &= \frac{(p_{\tilde{S}_{21}}p_{\tilde{S}_{22}}(1-p_{N_1}) + (1-p_{\tilde{S}_{21}})(1-p_{\tilde{S}_{22}})p_{N_1})(p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}}p_{\tilde{S}_{22}}(p_{\Theta_1} * p_B * p_{N_1}) + (1-p_{\tilde{S}_{21}})(1-p_{\tilde{S}_{22}})(1-p_{\Theta_1} * p_B * p_{N_1})}, \\
p_2 &= \frac{(p_{\tilde{S}_{21}}(1-p_{\tilde{S}_{22}})(1-p_{N_1}) + (1-p_{\tilde{S}_{21}})p_{\tilde{S}_{22}}p_{N_1})(p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}}(1-p_{\tilde{S}_{22}})(p_{\Theta_1} * p_B * p_{N_1}) + (1-p_{\tilde{S}_{21}})p_{\tilde{S}_{22}}(1-p_{\Theta_1} * p_B * p_{N_1})}, \\
p_3 &= \frac{(p_{\tilde{S}_{21}}p_{\tilde{S}_{22}}(1-p_{N_1}) + (1-p_{\tilde{S}_{21}})(1-p_{\tilde{S}_{22}})p_{N_1})(1-p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}}p_{\tilde{S}_{22}}(1-p_{\Theta_1} * p_B * p_{N_1}) + (1-p_{\tilde{S}_{21}})(1-p_{\tilde{S}_{22}})(p_{\Theta_1} * p_B * p_{N_1})}, \\
p_4 &= \frac{(p_{\tilde{S}_{21}}(1-p_{\tilde{S}_{22}})(1-p_{N_1}) + (1-p_{\tilde{S}_{21}})p_{\tilde{S}_{22}}p_{N_1})(1-p_{\Theta_1} * p_B)}{p_{\tilde{S}_{21}}(1-p_{\tilde{S}_{22}})(1-p_{\Theta_1} * p_B * p_{N_1}) + (1-p_{\tilde{S}_{21}})p_{\tilde{S}_{22}}(p_{\Theta_1} * p_B * p_{N_1})}. \tag{96}
\end{aligned}$$

$$\begin{aligned}
& -H(S_1) - H(N_1 \oplus N_2) \\
& = H_2(p_{\Theta_1} * p_{S_1}) + H_2(p_{\Theta_2} * p_{N_1} * p_{N_2}) \\
& \quad - H_2(p_{S_1}) - H_2(p_{N_1} * p_{N_2}). \tag{88}
\end{aligned}$$

Calculation of term $I(X_1X_2; Y|U)$: There is

$$\begin{aligned}
I(X_1X_2; Y|U) &= I(X_1X_2; Y|UU_2) \\
&= H_2(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1}) + H_2(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\
& \quad - H_2(p_{S_1}) - H_2(p_{S_{21}} * p_{S_{22}}). \tag{89}
\end{aligned}$$

Calculation of term $I(X_1X_2; Y)$: The detailed calculation is given as follows.

$$\begin{aligned}
I(X_1X_2; Y) &= H(Y) - H(Y|X_1X_2) \\
&= H(Y) - H(S_1) - H(S_{\text{comb}}) \\
&= H(U \oplus \Sigma_1 \oplus \Theta_1 \oplus S_1, U \oplus \Theta_2 \oplus S_{21} \oplus S_{22}) \\
& \quad - H(S_1) - H(S_{\text{comb}}) \\
&= H_4(b_1, b_2, b_3, b_4) - H_2(p_{S_1}) - H_2(p_{S_{21}} * p_{S_{22}}), \tag{90}
\end{aligned}$$

where

$$\begin{aligned}
b_1 &= p_U(1-p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1-p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\
& \quad + (1-p_U)(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}), \\
b_2 &= p_U(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1-p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\
& \quad + (1-p_U)(1-p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}), \\
b_3 &= p_U(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\
& \quad + (1-p_U)(1-p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1-p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}), \\
b_4 &= p_U(1-p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}) \\
& \quad + (1-p_U)(p_{\Sigma_1} * p_{\Theta_1} * p_{S_1})(1-p_{\Theta_2} * p_{S_{21}} * p_{S_{22}}). \tag{91}
\end{aligned}$$

APPENDIX G

CALCULATION OF TERMS IN DISTORTION RESULT OF ACHIEVABLE SCHEME IN [9]

Since the sensing state of transmitter 2

$$S_{T_2} = S_{21}, \tag{92}$$

the state estimator of transmitter 2 is

$$\begin{aligned}
\hat{S}_{21}(X_2, Z_2, U_1, V_1) &= \arg \min_{s'_{21} \in \{0,1\}} \\
& \sum_{s'_{21}} P_{S_{21}|X_2Z_2U_1V_1}(s_{21}|x_2z_2u_1v_1)d(s_{21}, s'_{21}). \tag{93}
\end{aligned}$$

Let $Z'_2 = \Theta_1 \oplus B \oplus \tilde{S}_{21} \oplus N_1$. It is easy to obtain that

$$\hat{S}_{21}(X_2, Z_2, U_1, V_1) = \hat{S}_{21}(Z'_2, V_1) \triangleq \arg \min_{s'_{21} \in \{0,1\}}$$

$$\sum_{s'_{21}} P_{S_{21}|Z'_2V_1}(s_{21}|z'_2v_1)d(s_{21}, s'_{21}). \tag{94}$$

Then, it is quite obvious that

$$\begin{aligned}
\hat{S}_{21}(Z'_2 = 0, V_1 = 0) &= \mathbb{1}\{p_1 \geq 0.5\}, \\
\hat{S}_{21}(Z'_2 = 0, V_1 = 1) &= \mathbb{1}\{p_2 \geq 0.5\}, \\
\hat{S}_{21}(Z'_2 = 1, V_1 = 0) &= \mathbb{1}\{p_3 \geq 0.5\}, \\
\hat{S}_{21}(Z'_2 = 1, V_1 = 1) &= \mathbb{1}\{p_4 \geq 0.5\}, \tag{95}
\end{aligned}$$

where p_1, p_2, p_3, p_4 are given in (96).

Then, the expected distortion D_2 over all Z'_2, V_1 based on the state estimator in (95) is given by

$$\begin{aligned}
D_2 &= P_{Z'_2V_1}(0, 0) \min\{p_1, 1-p_1\} \\
& \quad + P_{Z'_2V_1}(0, 1) \min\{p_2, 1-p_2\} \\
& \quad + P_{Z'_2V_1}(1, 0) \min\{p_3, 1-p_3\} \\
& \quad + P_{Z'_2V_1}(1, 1) \min\{p_4, 1-p_4\}. \tag{97}
\end{aligned}$$